DENVER INTERNATIONAL AIRPORT

STANDARDS MANUAL

Operational Readiness, Activation, and Transition (ORAT)

Design, Engineering and Construction (DEC)

Revised: Q4 2024



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Table of Contents

Table of Contents 3
Table of Figures 7
Table of Tables 8
Summary of Revisions 9
Purpose of Design Standards Manuals (DSM) 11
1.0.1. ORAT Standards Manual (OSM) 11
Variance from DEN Standards Manuals 11
Chapter 1- ORAT Introduction 1
1.0. Introduction and Purpose 1
1.0.1. Operational Readiness, Activation, and Transition (ORAT) 1
1.1. Intended Users 1
1.2. Vision 100 and Operation 2045 2
1.3. OSM Structure 3
1.3.1. Contents 3
1.3.2. DEN ORAT Roadmap 4
1.3.3. Project Categorization – Applying ORAT at Different Scales 5
1.3.4. Roles and Responsibilities 5
Chapter 2- Define Phase 7
2.0. Introduction 7
2.1. Understand Project Tier & ORAT Needs 8
2.1.1. Step 1: Understand Project 8
2.1.2. Step 2: Understand Project Tier 8
2.1.3. Step 3: Determine High-Level ORAT Needs 9
2.1.4. Successful Outcomes 9
2.2. Identify Impacted Stakeholders 10
2.2.1. Step 1: ORAT Input into Stakeholder Identification 10
2.2.2. Step 2: Document Stakeholders 10
2.2.3. Successful Outcomes 10
2.3. Develop Stakeholder Engagement & Facilitation Strategy 11
2.3.1. Step 1: ORAT Input into Stakeholder Mapping 11
2.3.2. Step 2: ORAT Input into Engagement & Facilitation Strategy 11
2.3.3. Step 3: Ongoing Stakeholder Status Check-Ins 12
2.3.4. Successful Outcomes 12
2.4. Initiate Operations & Maintenance Budget Forecast 13
2.4.1. Step 1: Begin Whole-Life Cost Estimates 13
2.4.2. Step 2: Define Information Requirements for the Contractor 13
2.4.3. Successful Outcomes 13
2.5. Produce ORAT Plan & Go-Live Milestone Schedule 14
2.5.1. Step 1: Produce Draft ORAT Plan 14
2.5.2. Step 2: Produce Go-Live Milestone Schedule 15

2.5.3. Successful Outcomes 15
2.6. Develop Risk Management Strategy 16
2.6.1. Step 1: ORAT Input into Project Risk Management Strategy 16
2.6.2. Successful Outcomes 16
2.7. Confirm ORAT Staffing and Support, Budget and Task Management 17
2.7.1. Step 1: Establish ORAT Team 17
2.7.2. Tier III 17
2.7.3. Tier II 17
2.7.4. Tier I 17
2.7.5. Step 2: Finalize ORAT Budget 17
2.7.6. Step 3: Establish ORAT Task Management 18
2.7.7. Successful Outcomes: 18
2.8. Apply Appropriate ORAT Governance & Reporting 19
2.8.1. Step 1: Establish the project within the existing OSC 20
2.8.2. Step 2: Establish the ORAT Management Group 20
2.8.3. Tier III 20
2.8.4. Tier II 20
2.8.5. Tier I 20
2.8.6. Step 3: Establish ORAT Reporting 20
2.8.7. Successful Outcomes 21
2.9. Define Phase Toolkit 22
Chapter 3- Deliver Phase 23
3.0. Prepare 23
3.0.1. Introduction 23
4.1 Operational Charrette 24
3.0.2. Step 1: Schedule Operational Charrette 24
3.0.3. Step 2: Deliver Operational Charrette 24
3.0.4. Tier III 24
3.0.5. Tier II 24
3.0.6. Tier I 24
3.0.7. Step 3: Develop Operational Requirements 25
3.0.8. Step 4: Prioritize and Select Operational Requirements 25
3.0.9. Successful Outcomes: 25
3.1. Develop Detailed ORAT Plan & Activation Requirements 26
3.1.1. Step 1: Develop Detailed ORAT Plan 26
3.1.2. Step 2: Develop Activation Requirements 27
3.1.3. Successful Outcomes: 27
3.2. Review Operational Impacts 28
3.2.1. Step 1: Conduct Operational Impact Assessment Workshop(s) 28
3.2.2. Tier III 28
3.2.3. Tier II 28
3.2.4. Tier I 28
3.2.5. Successful Outcomes: 29

3.3. Define System & Asset Responsibilities 30
3.3.1. Step 1: Develop Owner, Operator, Maintainer, User (OOMU) Responsibilities Matrix 30
3.3.2. Successful Outcomes: 30
3.4. Develop Detailed Operations & Maintenance and Asset Management Plans 31
3.4.1. Step 1: Develop Draft Operations and Maintenance Plans and Asset Management Plans 31
3.4.2. Step 2: Update Service Contracts 31
3.4.3. Successful Outcomes: 31
3.5. Deliver Phase: Prepare Toolkit 32
Chapter 4- Deliver Phase 33
4.0. Plan 33
4.0.1. Introduction 33
4.1. Identify & Develop SOPs and Contingency Plans 34
4.1.1. Step 1: Identify SOPs and Contingency Plans 34
4.1.2. Step 2: Develop or Update SOPs 34
4.1.3. Successful Outcomes: 34
4.2. Determine People Needs & Prepare Training & Familiarization Materials 35
4.2.1. Step 1: Conduct Training Needs Analysis 35
4.2.2. Step 2: Develop Training & Familiarization Schedule 35
4.2.3. Step 3: Support Technical Training 36
4.2.4. Step 4: Prepare Training & Familiarization Materials 36
4.2.5. Successful Outcomes: 37
4.3. Agree & Design Operational Exercises 38
4.3.1. Step 1: Identify Exercise Needs 38
4.3.2. Tier III 39
4.3.3. Tier II 39
4.3.4. Tier I 39
4.3.5. Step 2: Design Operational Exercises 39
4.3.6. Step 3: Schedule Operational Exercises 39
4.3.7. Successful Outcomes: 40
4.4. Develop Transition and Cutover Plan with Commissioning alignment 41
4.4.1. Step 1: Develop Transition and Cutover Plan 41
4.4.2. Step 2: Develop Transition Checklists and Review 42
4.4.3. Successful Outcomes: 42
4.5. Develop Integrated Schedule 43
4.5.1. Step 1: Develop ORAT Schedule 43
4.5.2. Step 2: Align ORAT Schedule in Program Schedule 43
4.5.3. Successful Outcomes: 43
4.6. Deliver Phase: Plan Toolkit 44
Chapter 5- Deliver Phase 45
5.0. Execute 45
5.0.1. Introduction 45
5.1. Track Asset & System Activation and Handover 46
5.1.1. Step 1: Develop the Handover Plan/Schedule for Assets and Systems 46

5.1.2. Step 2: Track Activation and Handover 46
5.1.3. Successful Outcomes: 46
5.2. Collect Asset Information, Critical Spares & Prepare Maintenance Plans 47
5.2.1. Step 1: Collect Asset Information 47
5.2.2. Step 2: Acquire Critical Spares List 47
5.2.3. Step 3: Prepare Maintenance Plans 47
5.2.4. Successful Outcomes: 48
5.3. Deliver Training 49
5.3.1. Step 1: Coordinate and Execute Training and Familiarization 49
5.3.2. Step 2: Track People Readiness Progress 49
5.3.3. Successful Outcomes: 49
5.4. Deliver Operational Exercises 50
5.4.1. Step 1: Develop Controller and Evaluator Handbook 50
5.4.2. Step 2: Monitor Status of all Dependencies 50
5.4.3. Step 3: Schedule a Go/No-Go with the OSC 50
5.4.4. Step 4: Execute Operational Exercises 50
5.4.5. Step 5: Collect Observer Feedback 51
5.4.6. Step 6: Compile an Exercise After Action Review (AAR) and record issues raised by Operational Exercise in the Issues Tracker 51
5.4.7. Successful Outcomes: 51
5.5. Mobilizations & Space Handover 52
5.5.1. Step 1: Onboard Move Coordinator and Prepare Moves 52
5.5.2. Step 2: Execute Space Handover and Moves 52
5.5.3. Successful Outcomes: 52
5.6. Transition & Cutover 52
5.6.1. Step 1: Execute various Transition Checklist Items 53
5.6.2. Step 2: Schedule a Go/No-Go with the OSC 53
5.6.3. Step 4: Identify post-opening support requirements 54
5.6.4. Successful Outcomes: 54
4.2 Deliver Phase: Execute Toolkit 55
Chapter 6- Closeout Phase 57
6.0. Introduction 57
6.1. Issue Management & Resolution 58
6.1.1. Step 1: Update ORAT Issues Tracker 58
6.1.2. Successful Outcomes: 58
6.2. Conduct Lessons Observed Review 59
6.2.1. Step 1: ORAT Support in a Lessons Observed Survey and/or Workshop 59
6.2.2. Successful Outcomes: 59
4.3 Close Out Phase: Toolkit 59

Table of Figures

Figure 1-1: DEN ORAT Roadmap *4*

Figure 2-1: Influence and Interest Matrix *11*

Table of Tables

Table 4-2: Types of Operational Exercises *38*

Table 4-3: Types of Opening Strategies *41*

Summary of Revisions

Fourth Quarter

Reference	Revision Description
ORAT Standards Manual	Created new manual

Revision Notation: Revisions made to this Manual during this revision cycle are annotated as shown in the example below:

A vertical line in the left-hand margin is used to annotate paragraphs that have been added or revised in the current publication. Revisions may include items such as new requirements, clarification of existing requirements, or removal of requirements that no longer apply to projects. Revision annotation is applied to each publication individually; revisions made in past publications are not annotated in subsequent publications.

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Purpose of Design Standards Manuals (DSM)

The DEN Design Standards have been developed to ensure a unified and consistent approach to the thematic and technical design for DEN. These standards are for use and strict implementation by all consultants under contract to DEN, to tenants, and all other consultants under contract to any other entity for the design of projects at DEN. The Standards Manuals are working documents, which are revised and updated, as required, to address the general, conceptual, design, and technical standards for all areas of design for DEN.

1.0.1. ORAT Standards Manual (OSM)

This ORAT Standards Manual (OSM) for DEN has been prepared for use by competent, professionally licensed architectural and engineering consultants under the direction of DEN Design, Engineering and Construction (DEC) or tenants of DEN. The OSM has been developed using industry best practices and through internal DEN stakeholder workshops to ensure it reflects the DEN Project Life Cycle (PLC) and environment. The approach should be used as a framework, and it is expected to be customized to fit the needs of the specific project and stakeholders.

The OSM is intended to bring a wide range of resources together in one place to help project teams successfully achieve airport and stakeholder goals and objectives. It provides a framework to deliver fully functional facilities, on-time project handover, mitigated opening days and project activations, and prepared stakeholders for initial and ongoing operations. Every DEN project is unique; some may be large in scale while others are small with a greater impact on operations. All projects, however, should follow a considered delivery approach to ensure project objectives are achieved and the OSM provides the structure to enable that goal. By embedding ORAT into the PLC, we create a foundation for a well-prepared and efficiently operating airport, ready to meet the demands of modern air travel. The OSM outlines a delivery approach that should be used across all projects. It aims to provide consistency for all project types and stakeholders.

The OSM is intended to be used as a whole, as each manual is complementary to the other Design Standards Manuals (DSM). To understand the overall thematic and standards for DEN, the applicable manuals must be utilized together and not separated from the DSMs. The standards shall not be quoted, copied, or referenced in any bidding or construction contract documents. Content contained in the OSM shall not be copied in any bidding or construction documents, except where specifically instructed to do so. All information contained in these standards must be fully explained and shown in all bidding and contract documents. The Consultant shall not reproduce, duplicate in any manner, transmit to other consultants or other entities, or use in conjunction with other projects without the express written consent of DEN.

The OSM is a “live” document which will be reviewed, developed, and updated on a regular basis to ensure that it captures lessons learned from future pilots and projects. Any requests for updates and changes should be directed to the DEC ORAT Supervisor for consideration.

NOTE: This document is optimized for duplex (double-sided) printing.

Variance from DEN Standards Manuals

Requests for non-conformance or variance from DEN Standards manuals, for any DEN or Tenant Projects, must be formally submitted using the online manual Variance Request form at the following website:



DEN DSM Variance Request Form

Variance requests may or may not be approved by DEN and response will be communicated to the requestor.

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Chapter 1- ORAT Introduction

1.0. Introduction and Purpose

1.0.1. Operational Readiness, Activation, and Transition (ORAT)

ORAT is a systematic approach used to ensure that new or significantly renovated facilities are fully prepared to operate efficiently and effectively from day one. ORAT focuses on coordinating the efforts of various stakeholders, testing systems and procedures, and training staff before the facility opens to the public.

A. Key Objectives of ORATSeamless Operations: Ensuring all airport operations run smoothly.

B. Staff Preparedness: Training airport staff to handle new equipment, systems, and procedures effectively.

C. System Integration: Testing and validating that all technological systems, such as IT networks, communication systems, and security protocols, are fully integrated and functional.

D. Stakeholder Coordination: Bringing together airlines, airport authorities, service providers, and other stakeholders to collaborate and align on operational procedures.

E. Transparent Information: providing real-time information on the state of readiness to gain stakeholder confidence and align expectations.

F. Passenger Experience: Ensuring the facility is ready to provide a positive and efficient experience for passengers from the moment they enter the airport.

The DEN ORAT approach concentrates on complete readiness, not just operational readiness. Our approach is all encompassing and includes preparations to open, operate, manage, and maintain a new service. This approach facilitates the opportunity to make progressive changes and adjustments ahead of the formal opening to assure success.

1.1. Intended Users

The intended users of the OSM are those who will be involved in ORAT delivery on projects, however, an awareness of ORAT processes for any project role or stakeholder at DEN will be beneficial.

Users include:

1. ORAT Project Lead – this is the dedicated DEN ORAT resource that will manage ORAT implementation across projects.

2. ORAT Team – depending on the project, a dedicated ORAT team may be established to lead the coordination of ORAT activities on projects.

3. Project PMs – across all project types, the PMs will need to have an awareness of the ORAT requirements and how it impacts their projects. For smaller projects, direct ORAT Team support may be limited. Therefore, the PMs will need to have information to ensure ORAT will be incorporated into their projects.

4. DEN Operations, Maintenance, and Asset Management Groups – stakeholders in these groups will be the beneficiaries and key collaborators in the ORAT process as it aims to provide a smooth handover to these groups for business-as-usual.

1.2. Vision 100 and Operation 2045

Vision 100 and Operation 2045 are two phases of DEN's strategic plan. Vision 100 is phase one and is focused on preparing the airport to serve 100 million annual passengers in the next several years. Operation 2045 is phase two and is focused on preparing the airport for its 50th anniversary in 2045 and for an expected 120 million-plus annual passengers. Both phases combined serve as a blueprint to align decision-making and accountability. Learn more on flydenver.com. The ORAT process supports Vision 100 and Operation 2045 in the following manner:

Sustainability and Resilience: ORAT strengthens both resiliency and environmental sustainability in airport operations. By carefully planning for new facilities and systems, ORAT ensures they can adapt to disruptions, enhancing operational resilience. Additionally, ORAT incorporates sustainable practices, emphasizing energy efficiency, waste reduction, and resource conservation in new operations. This approach supports the airport's long-term sustainability goals and aligns developments with broader environmental commitments.

EDI & Accessibility: At its core, ORAT is committed to comprehensive stakeholder engagement, ensuring that a broad range of perspectives are included. This involves engaging not only with various divisions and sections within DEN but also with external stakeholders, including airlines, tenants, and federal partners. ORAT has adopted a proactive approach by collaborating closely with a variety of stakeholders with diverse viewpoints from ADA expertise to customer experience and more beginning from the Define phase of projects. This collaboration ensures that accessibility considerations are integrated early on, facilitating connections with DECs project managers to guarantee that all individuals passing through DEN have an inclusive and accessible experience.

Operational Excellence: ORAT is the epitome of Vision 100's Operational Excellence guiding principle. The core value of ORAT is functionality for the end user. This is accomplished through close collaboration with partners DEN wide to ensure a smooth transition from project definition to delivery and closeout. ORAT also supports project lessons learned, which allows DEN to identify critical touchpoints that should be accounted for in subsequent initiatives to mitigate past issues and augment past successes.

Enhancing the Customer Experience: The individuals that pass-through DEN, including employees and passengers, are the primary customers of DEN projects. By consistently focusing on their future needs throughout the project lifecycle, ORAT aims to deliver a final product that feels seamlessly integrated, with minimal disruption or inconvenience. ORAT prioritizes these customers and actively advocates for needs, ensuring that input is incorporated into the project. This approach helps facilitate a smooth transition to daily operations.

1.3. OSM Structure

1.3.1. Contents

The OSM contains the following chapters:

Chapter 1 | ORAT Introduction. This chapter introduces ORAT, how the OSM aligns and supports Vision 100. It also includes the DEN ORAT Roadmap which succinctly highlights the ORAT activities across the DEN PLC. It also defines the key principles of scalability *and* the impact it has on how ORAT is applied to projects.

Chapter 2 | Define Phase. This chapter sets out each dedicated activity for the Define Phase which ensures early ORAT and stakeholder awareness of the project Business Case, developing a preliminary ORAT plan and establishing governance, management, and control of ORAT and associated activities. Step-by-step guidance is provided on how to enact the specific activity and its successful outcomes. Where applicable, it identifies how the activity changes based of the scale of the project. The chapter concludes with a Toolkit that identifies a Checklist of items and any available Templates or other useful resources available.

Chapter 3 |Deliver Phase: Prepare. This chapter sets out each dedicated activity for the Prepare stage of the Deliver Phase which ensures end-user input into the design; reviewing operational impacts with each identified stakeholder and initiating plan updates for Operations &Maintenance (O&M) and Asset Management. Step-by-step guidance is provided on how to enact the specific activity and its successful outcomes. Where applicable, it identifies how the activity changes based of the scale of the project. The chapter concludes with a Toolkit that identifies a Checklist of items and any available Templates or other useful resources available.

Chapter 4 |Deliver Phase: Plan. This chapter sets out each dedicated activity for the Plan stage of the Deliver Phase which is developing processes and procedures for use in the new facility. Identifying people needs along with designing operational exercises to prove that people, processes, systems, and assets work together. Step-by-step guidance is provided on how to enact the specific activity and its successful outcomes. Where applicable, it identifies how the activity changes based of the scale of the project. The chapter concludes with a Toolkit that identifies a Checklist of items and any available Templates or other useful resources available.

Chapter 5 |Deliver Phase: Execute. This chapter sets out each dedicated activity for the Execute stage of the Deliver Phase ensuring people are ready for transition in terms of training and familiarization; operational exercise deliver and management from a construction site into a live operational facility. Step-by-step guidance is provided on how to enact the specific activity and its successful outcomes. Where applicable, it identifies how the activity changes based of the scale of the project. The chapter concludes with a Toolkit that identifies a Checklist of items and any available Templates or other useful resources available.

Chapter 6 |Closeout Phase This chapter sets out each dedicated activity for the Closeout Phase ensuring issues are resolved and lessons learned for continuous improvement. Step-by-step guidance is provided on how to enact the specific activity and its successful outcomes. Where applicable, it identifies how the activity changes based of the scale of the project. The chapter concludes with a Toolkit that identifies a Checklist of items and any available Templates or other useful resources available.

1.3.2. DEN ORAT Roadmap

The DEN ORAT Roadmap, detailed below, is organized to align with DEN's PMO Project Life Cycle (PLC). It outlines specific activities and considerations that need to occur during the various phases of a project to assure that end state operations are at the forefront throughout project delivery. Not all tools or level of effort described in the OSM are appropriate for each project, so it is important that the Project Categorization (described in the next section) is understood early in the PLC so ORAT can be applied efficiently and effectively. In Figure 1, the numbering corresponds to the applicable section found within the OSM for ease of navigation.

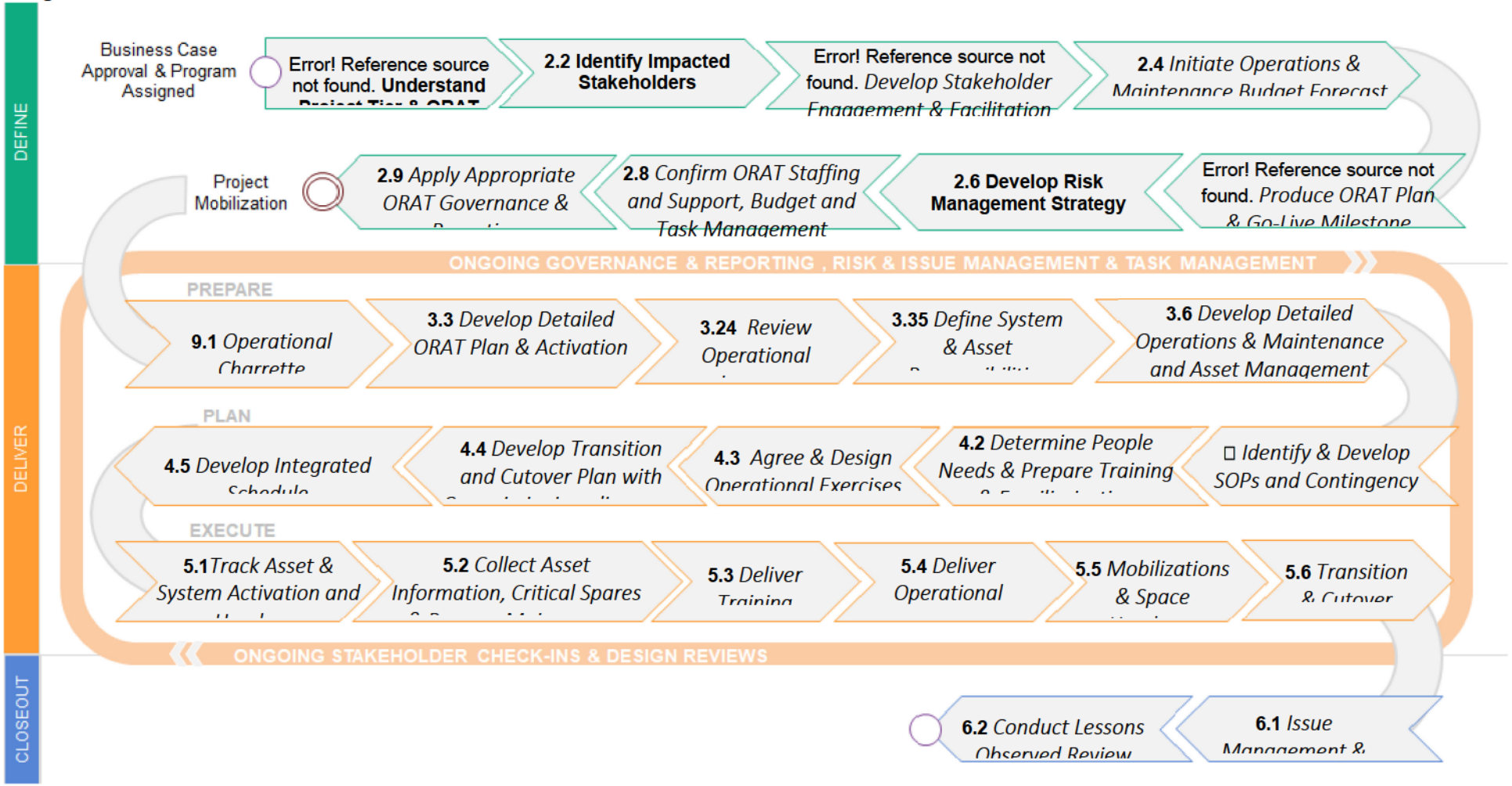


Figure 1-1: DEN ORAT Roadmap

1.3.3. Project Categorization – Applying ORAT at Different Scales

ORAT provides a framework for ensuring end-user operators and maintainers are considered throughout the design process and the transition from construction into operation. ORAT activities, such as an Operational Impact Assessment and Stakeholder Engagement, apply to every DEN project, large or small, to ensure that nothing is overlooked.

What varies per project is the scale of each activity – for example, how many stakeholders are engaged with, how many operational planning workshops are held, how many communications are sent, or how many Operational Exercises are conducted.

Categorization systems are critical for describing, organizing, and managing the project portfolio within large organizations. Using a categorization model based on project complexity, risk, and value, provides DEN with improved visibility of the capital portfolio and the ability to measure key metrics for resourcing and portfolio planning.

The OSM provides guidance on how to scale ORAT activities up to fit larger projects and down to fit smaller projects throughout all sections depending on the Tier assigned following the categorization of the project which occurs in the PLC Initiation Phase.

1.3.3.1. Project Tier Definition

DEN uses a project categorization system that breaks the portfolio into three (3) groups: Tier I, Tier II, and Tier III projects. The category (Tier) assigned to the project determines which level of governance it requires. It also helps inform the planning and execution of ORAT activities.

The project category is determined by evaluating ten (10) important aspects of each project. The category (Tier I, Tier II, or Tier III) is based on categorization scores, driven by a scoring table using options weightings to assign a maximum of a hundred points across the ten aspects. The aspects assessed include:

1. Project Type
2. Risk
3. Budget
4. Duration
5. Project Maturity/Complexity
6. Dependencies
7. Stakeholders
8. Operation, Safety, & Security
9. Impact to Customer Experience
10. Project Deadline Constraints

For further details, please refer to [Project Categorization Methodology].

1.3.3.2. Application Principles

In the Define Phase, the Project Tier should be understood within the Understand Project Tier & ORAT Needs phase of the project and considered in the development of the ORAT Plan & Go-Live Milestone Schedule.

For the Deliver Phase, where applicable, certain sections in the OSM contain guidance on how ORAT should be scaled for different Project Tiers. As a guiding principle, all ORAT activities should be conducted for every project scale. The OSM should be used to inform how many and to what extent activities should be carried out.

1.3.4. Roles and Responsibilities

ORAT is a collaborative process that will be led by the DEC ORAT Supervisor, a role residing in DEN DEC. The

ORAT Supervisor is responsible for the successful implementation and delivery of the DEN ORAT Approach across DEN projects. Project Managers will be close partners with the ORAT Project Lead to enable the integration of ORAT into all projects, regardless of size and scale.

Operations, Maintenance, and Asset Management are key stakeholders in an ORAT Program as these groups are the end-users of a project's assets, systems, and spaces, and therefore, who the ORAT Program is serving through the facilitation of a smooth transition into business-as-usual.

For each activity detailed in the OSM, the group with Primary Responsibility for delivery is indicated, as well as the Key Collaborator(s). A sample table is shown on the right. While the ORAT Project Lead is responsible for overall ORAT outcomes, they are not responsible for each activity. The ORAT Project Lead should provide the framework that enables collaborators in their readiness activities.

Primary Responsibility	ORAT
Key Collaborator(s)	Ops, Mx, PM
Project Tier Applicability	All

The following groups are referred to throughout the document:

- Project Sponsor (Sponsor) - responsible for overall project definition and preliminary budget in the Define Phase. The Project Sponsor is consulted by the Project Manager in the remaining phases through Closeout.
- Project Management Office (PMO) – responsible during the Define Phase before the project has been mobilized.
- Project Manager (PM) - responsible for overall project delivery from Project Mobilization to Project Close Out. They enable the ORAT Project Lead by providing information to support ORAT implementation and should incorporate ORAT into project schedules and budgets.
- DEC ORAT Supervisor (ORAT Supervisor) – a role residing in DEN DEC. The ORAT Supervisor is responsible for the successful implementation and delivery of the DEN ORAT Approach across DEN projects.
- ORAT Project Lead (ORAT) – assigned party by the DEC ORAT Supervisor and is responsible for successful delivery of the ORAT program and its successful outcomes. Facilitates and supports collaboration across other stakeholder groups to ensure readiness activities are completed.
- Operations, Maintenance and Asset Management Groups (Ops, Mx and AM) – responsible for their own readiness by engaging in the ORAT process and ensuring the timely completion of activities, facilitated by the ORAT Project Lead.

End of Chapter

Chapter 2- Define Phase

2.0. Introduction

Purpose: To set the ORAT project / program up for success and to establish the basis for the management and control.

Considerations:

A. Who are all the impacted stakeholder groups?

B. How will the project / program be managed and controlled?

C. What ORAT governance structures need to be in place?

D. How do we ensure alignment on the future end state?

E. What does Opening Day look like, can we smooth or soften the opening by understanding what is critical and not critical?

F. How do we ensure the broader stakeholder group is clear on our approach?

G. Who do we need to engage with and communicate to, and with what method and frequency?

H. How do we ensure high-level ORAT risks and issues are identified early and discussed regularly?

2.1. Understand Project Tier & ORAT Needs

This activity focuses on sufficiently understanding the project context, its size and its impact, to determine high-level ORAT Needs, such as resources and budget. The following approach will be used to ensure the Project Sponsor and ORAT Project Lead are aligned in their understanding of the project and requirements for ORAT.

Primary Responsibility	ORAT
Key Collaborator(s)	Sponsor
Project Tier Applicability	All

2.1.1. Step 1: Understand Project

The project has not yet been mobilized but its intent should have been defined through the development of the approved Business Case. At this stage, the ORAT Project Lead should familiarize themselves with the approved Business Case, the future project's benefits and risks, and its scope, if known. This will enable the ORAT Project Lead to make informed decisions on the ORAT Needs of the future project.

2.1.2. Step 2: Understand Project Tier

The categorization of a project takes place during the PLC Initiation Phase by the PMO, following the criteria outlined in the DEN PLC Project Categorization Methodology document. This categorization, or Tier, determines the level of effort and application for project delivery in the broader context. It should also reflect operational impact and the extent of ORAT activities needed.

The operational impact of the project considers how much end-users will be impacted by the project outcome. For example, if a new technology is being introduced at DEN that has not been previously used, users, operators and maintainers will need to undergo extensive training, testing, and documentation review, more so than with a familiar system replacement. Additionally, the project may require a new process to be developed or new job roles to be created to operate following Cutover and go-live.

As mentioned earlier, the Tier is determined by evaluating ten (10) key aspects. By considering the following statements alongside the Project Categorization, you can better assess the ORAT Needs, as different levels of ORAT application are required based on these factors. The more statements that are true, the greater operational impact.

A. A new operational process will need to be developed.

B. Technology or equipment is being implemented that has not been used, operated, or maintained by end-users.

C. There is significant difference in geographical location or facility layout.

D. An entirely new job role or job function need to be created.

E. The project involves or requires endorsement with 5+ stakeholders.

This initial consideration is high-level at this stage. A more robust Operational Impact Assessment should be conducted during the Deliver Phase: Prepare.

Throughout applicable subsequent sections in the OSM an ORAT application for the different Project Tiers is provided. This provides guidance on what ORAT activities should be applied depending on the size of the project and its operational impact. As a guiding principle, **all** ORAT activities should be conducted for every Project Tier, the OSM should be used to inform how many and to what extent activities should be carried out.

2.1.3. Step 3: Determine High-Level ORAT Needs

The ORAT Project Lead should determine the High-Level ORAT Needs based on the Project Tier designation. This early understanding of the ORAT Needs allows for sufficient time to mobilize required resources (e.g., labor and/or budget) and to provide awareness of requirements that may be asked of the project (e.g., time allocation in project schedule for delivering ORAT activitiesTypes of ORAT Needs that should be considered are:

- Will operational processes and Standard Operating Procedures (SOPs) need to be redefined? Will Operations & Maintenance and Asset Management Plans need to be updated?
- Will operations need to be exercised due to the level or complexity of change or impact the project has on the facility, process, or staff? Will time need to be dedicated in the project schedule for these Operational Exercises?
- Will training need to be delivered? Will this likely be delivered by suppliers, or will supplementary training need to be developed by an ORAT Team?
- What ORAT resources will be needed to deliver these ORAT activities? Will additional labor be needed (e.g., external consultants) and will a budget be required?

These Needs are intended to only be high-level at this stage for the purpose of providing early awareness to inform the mobilization of the project. These Needs should be documented for reference later as they will be pulled into the ORAT Plan, in the Project Management Plan (PMP) as defined by the DEC PMO.

2.1.4. Successful Outcomes

- ✓ **Clarity on the Project Tier** – its size and impact on stakeholders.
- ✓ **Understanding of its ORAT Needs** – whether an ORAT Team will be required to deliver readiness activities such as training and operational exercises.

2.2. Identify Impacted Stakeholders

Primary Responsibility	Sponsor
Key Collaborator(s)	ORAT
Project Tier Applicability	All

Impacted stakeholders are defined as all parties with an interest in, or influence (positive or negative) on the successful activation and operation of the project. Impacted stakeholders includes anyone that has operations in the space based on the specific project. Identifying all stakeholders is crucial to the success of the project to generate buy-in, ensure awareness and preparedness, and mitigate resistance. This activity is a precursor to developing the Stakeholder Engagement and Facilitation Strategy, as part of the PMP.

2.2.1. Step 1: ORAT Input into Stakeholder Identification

The ORAT Project Lead should work with the Project Sponsor and internal teams to identify and document stakeholders using the Business Case as a key reference document. The identification should be conducted holistically – considering operations, maintenance, and management roles.

2.2.2. Step 2: Document Stakeholders

A comprehensive contact list documenting project stakeholders and their respective single point of contacts, and notes from engagement should be developed and maintained by the project with input from, and accessible to, the ORAT Project Lead. The key stakeholders should be notified of the project at this stage for awareness and initial engagement. This will be further developed throughout the project and used throughout to monitor engagement.

Standard DEN stakeholder groups include, but are not limited to airlines, airline service providers, baggage and ground service operations groups, building operations and facilities management groups, janitorial groups, TSA, security services, IT, and emergency responders.

2.2.3. Successful Outcomes

- ✓ **Stakeholders identified** – all stakeholders impacted by the future project are identified.
- ✓ **Stakeholder contacts documented** – points of contact for impacted stakeholders are documented and notified of project.

2.3. Develop Stakeholder Engagement & Facilitation Strategy

Primary Responsibility	Sponsor
Key Collaborator(s)	ORAT
Project Tier Applicability	All

It is important to engage with the right people, at the right level, at the right time. Therefore, once stakeholders have been identified, a mapping process, supported by the ORAT Project Lead, will be undertaken to inform an effective Stakeholder Engagement and Facilitation Strategy which will be included in the PMP. A robust stakeholder engagement approach should be implemented that is clear, consistent, and effective at getting messages across to impacted parties.

2.3.1. Step 1: ORAT Input into Stakeholder Mapping

The ORAT Project Lead supports the identification of stakeholders—all parties with an interest in, or influence (positive or negative) on, the successful activation and operation of the project—on a map of influence and interest.

This mapping results in stakeholders being assigned into a relevant quadrant that informs the level of management and communication required.

Throughout the project, the ORAT Project Lead, with the Project Sponsor, should continue to analyze project stakeholders and monitor each stakeholder's influence and interest to facilitate the most effective method of communication.

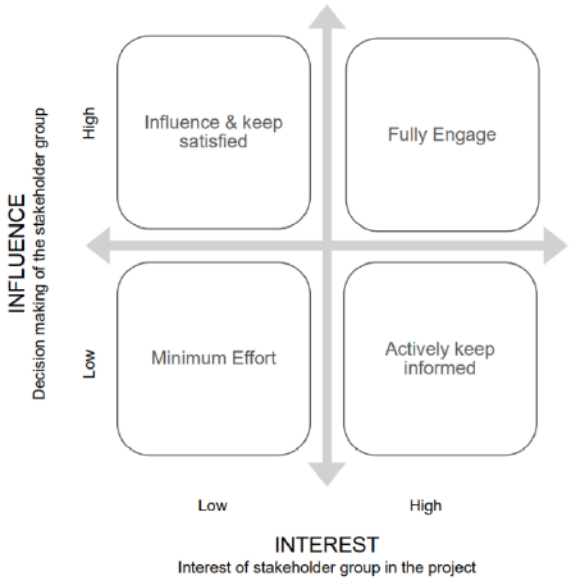


Figure 2-1: Influence and Interest Matrix

2.3.2. Step 2: ORAT Input into Engagement & Facilitation Strategy

Once stakeholders have been mapped, the ORAT Project Lead should support the development of the level of engagement, management and communication for each stakeholder group based on the quadrant they were allocated to. Approaches for consideration include:

A. Existing Project Management and Coordination Meetings

B. Fortnightly/weekly 1:1 Check-Ins for two-way communication

C. Monthly All-Stakeholder Conference Calls

D. Monthly Email Update Communications

E. Reporting Dashboard Access

Guiding Principles for communication should also be documented in the Strategy. They could include:

A. Openness and honesty are vital;

B. Comprehensiveness is important;

C. Relevance is a key priority; people should not be burdened with information which has no relevance to them, nor should it be more detailed than is necessary;

D. Timeliness is crucial; information must be up-to-date and available when it is needed; and

E. Plain language must be used, acronym-free and jargon-free and with further explanation where needed.

2.3.3. Step 3: Ongoing Stakeholder Status Check-Ins

As operational readiness transitions from the Define to Deliver Phase, periodic status check-ins with stakeholders are utilized to monitor progress of readiness activities and keep the project on track. This is represented by the orange circle on the DEN ORAT Roadmap around the Deliver Phase activities.

Stakeholders can be monitored by reviewing progress of readiness activities, such as updating or defining new SOPs, completion of Training & Familiarization activities, participation in Operational Exercises and finally their Transition activities. Additionally, 1:1 stakeholder check-ins offer the ORAT Team an opportunity to receive feedback and gauge stakeholders' perceived level of confidence for Opening Day.

2.3.4. Successful Outcomes

- ✓ **Stakeholders mapped** – stakeholders are mapped according to their influence and interest in the project to determine their engagement and communications needs.

- ✓ **Communication plan agreed** – using the stakeholder mapping, a robust communication plan should be agreed to ensure each group is informed to the right level at the right time.

2.4. Initiate Operations & Maintenance Budget Forecast

Primary Responsibility	AM, Ops, Mx
Key Collaborator(s)	ORAT
Project Tier Applicability	All

Early awareness of new systems and assets and their potential impact to resourcing and budgets provides significant value to operations, maintenance, and asset management groups. This information allows these groups to appropriately plan their future forecasts to ensure they can incorporate these assets into operations in a cost-effective and stable way. The following approach will be used to ensure stakeholders (e.g., operations, maintenance, facilities management) begin preparing budget forecasts before project mobilization considering preventative maintenance, replacements and upgrades, and staffing for operations and maintenance.

The ORAT Project Lead will support operations, maintenance, and asset management groups to complete these activities if needed.

2.4.1. Step 1: Begin Whole-Life Cost Estimates

At this stage of the Define Phase, impacted stakeholders (e.g., operations, maintenance, facilities management, asset management) should begin preparing whole-life asset cost estimates for operation and maintenance and relevant Business Case information should be provided to the stakeholders to enable them to update their existing estimation tools and templates. The whole life cost estimates serve to provide a more comprehensive understanding of the true cost of the asset (initial cost; operating cost; maintenance cost and end-of-life cost), the total cost of owning the asset over its entire lifespan. These will feed into future planning and initial budget forecasts that will need to be reevaluated after the design is further developed later in the project.

2.4.2. Step 2: Define Information Requirements for the Contractor

At this stage, Asset Management should consider defining documentation requirements for the contractor (e.g., asset warranty information, commissioning documentation, operations and maintenance (O&M) manuals, preventive maintenance schedules, asset data, etc.) that will allow them to prepare for the integration of any new assets. These requirements should not be technical but consider more the timelines and templates needed to ensure they are usable for future end-user groups.

2.4.3. Successful Outcomes

- ✓ **Whole-life cost estimates completed** – using the Business Case, early forecasts should be completed so operations, maintenance, and asset management groups can plan accordingly.

- ✓ **Contractor information requirements agreed** – clear requirements for the Contractor should be articulated and agreed to ensure the smooth handover and integration of assets later in the project.

2.5. Produce ORAT Plan & Go-Live Milestone Schedule

Primary Responsibility	Sponsor
Key Collaborator(s)	ORAT Sup
Project Tier Applicability	All

At this stage there is a lot of information that is still unknown and as such the Draft ORAT Plan articulates how ORAT is intended to be delivered for the project and is refined later when more details are known in the Deliver Phase. It describes the governance; management processes and readiness activities that could be implemented and their intended outcomes. The Plan intends to answer the who, what, where, why, how, and when of ensuring a successful go-live of the project. The Go-Live Milestone Schedule identifies the milestones that will be critical to the delivery of ORAT activities to provide early awareness of ORAT timescales to the Project Sponsor.

The ORAT Plan and Go-Live Milestone Schedule will be developed by the Project Sponsor with integral support from the ORAT Project Lead.

2.5.1. Step 1: Produce Draft ORAT Plan

While not all the necessary information is known at this point it is important to start considering what the answers might be across these different components, and the intention documented that will then be refined in the Deliver Phase. Using the ORAT Plan Template, the Project Sponsor should produce a Draft ORAT Plan outlining the anticipated approach to ensuring the readiness of operations given the current understanding of the impact of new facilities and/or assets and technologies. The Draft ORAT Plan should consider the following components:

A. ORAT Governance including risk management and reporting processes.
   a. What governance 'layers' need to be in place for the project? Will this project require additional working groups and reporting to the ORAT Steering Committee (OSC)?
   b. How will specific ORAT risks and issues be managed on the project?
   c. How will ORAT progress be reported and to whom?

B. Project Management including team roles and task management.
   a. What team roles and responsibilities are required, depending on the Project Tier?
   b. How will the team be managed, and tasks allocated?

C. Construction and Commissioning Alignment.
   a. What interfaces and dependencies are believed to be required with construction and commissioning activities, if any?
   b. How are the construction and commissioning teams going to be engaged with to ensure the right information is received at the right time?

D. Operations Planning
   a. Will operations and maintenance processes need to be reviewed and with which groups?
   b. Will existing plans and documents need updating?
   c. Will new plans and documents need to be developed?
   d. How long might this planning take?
   e. Will additional staff and resources be required?

E. Training & Familiarization
   a. Is training and familiarization anticipated to be required?
   b. What contractual training is anticipated to be delivered, if any?

F. Operational Exercises
   a. Are Operational Exercises likely to be required?
   b. What type (e.g., Drill; Functional; Full-scale or Emergency)?

G. Transition & Go-Live
    a. Is the project go-live anticipated to be phased?
    b. Are moves expected to be required?

2.5.2. Step 2: Produce Go-Live Milestone Schedule

The Go-Live Milestone Schedule is intended to be a high-level schedule outlining the timescales associated with the identified activities in the ORAT Plan e.g. Training and Familiarization and Operational Exercises.

At this stage, only a list of key milestone dates and a simple visual roadmap is required. Later, an Integrated Schedule will be developed with dependencies between construction and commissioning activities.

The Go-Live Milestone Schedule should be shared with the Project Sponsor once the project is mobilized to ensure early awareness of ORAT activities.

2.5.3. Successful Outcomes

- ✓ **ORAT Plan documented in PMP** – the ORAT Plan template is updated for the project and incorporated into the PMP.

- ✓ **Go-Live Milestone Schedule developed** – the key milestones and activities of the project, for the ORAT Plan, are visualized in a Schedule to provide early awareness of ORAT needs.

2.6. Develop Risk Management Strategy

Primary Responsibility	Sponsor
Key Collaborator(s)	ORAT
Project Tier Applicability	All

Managing and mitigating risk is key to the successful delivery of all projects. At DEN, risks are managed as part of the overall project - whether it be threats or opportunities to the budget and schedule. Additionally, risks associated with a successful operational readiness, activation, and transition to end-user groups are captured at the project level.

The ORAT Project Lead will provide input into the Risk Management Strategy for the project, including the early identification of risks, and throughout the course of the project.

2.6.1. Step 1: ORAT Input into Project Risk Management Strategy

The Project Sponsor will be drafting the Risk Management Strategy as part of the PMP, following PLC DSM guidelines. As part of this, the ORAT Project Lead should input risks to successful operational readiness, activation, and transition to end-user groups. These could include:

A. Lack of reasonable Program Schedule.

B. Lack of central reporting and awareness of key milestones.

C. Lack of dedicated time for testing.

D. System integration failure and timeliness of information provision.

Throughout the project, specific risks will arise that should be captured for mitigation actions to be agreed. The ORAT Project Lead will need to collaborate with the Project Sponsor throughout the project to manage risks. The Risk Management Strategy is ongoing throughout the project.

2.6.2. Successful Outcomes

- ✓ **ORAT Risks incorporated into Project Risk Management Strategy** – risks to successful operational readiness, activation, and transition are acknowledged by the Project Sponsor and included in the overall Risk Management Strategy.

2.7. Confirm ORAT Staffing and Support, Budget and Task Management

Primary Responsibility	ORAT
Key Collaborator(s)	Sponsor
Project Tier Applicability	Varies

Based on the Project Tier determined, the ORAT Project Lead may assign DEN resources or external resources to execute the ORAT Plan. The Confirm ORAT Staffing and Support, Budget and Task Management activity focuses on the assignment and onboarding of ORAT staff resources to the project and the setup of ORAT resource and task management to deliver ORAT activities. The following approach will be used to initiate the ORAT Team for the project.

2.7.1. Step 1: Establish ORAT Team

The ORAT Project Lead should appoint ORAT staffing and support to form the ORAT Team for the project based on the project tier and ORAT needs determined in the definition phase. The ORAT Project Lead is responsible for establishing the ORAT Management requirements for the project.

The ORAT Project Lead should:

A. Assign roles and responsibilities for all members of the ORAT Team and the roles and responsibilities of the ORAT Management Group members;

B. Communicate the organizational structure for the ORAT Team and extended team (e.g., ORAT Management Group);

C. Agree a risk identification process that clearly defines the methodology for raising risks to operational readiness and escalating them to the project;

D. Agree an issue resolution process that clearly defines the methodology for raising issues and resolving them;

E. Create a contact list for participants providing a single source for ORAT participants across all stakeholders;

F. Establish workstreams for the various ORAT tasks and activities. This is a key function in which the ORAT Team and stakeholders participate to generate the materials necessary to move the ORAT program forward; and

G. Ensure awareness of all statutory and regulatory requirements that need to be achieved to put an airport facility into operation. A key part of the success of any ORAT program is early identification of these requirements.

2.7.1.1. Application at different Project Tiers:

2.7.2. Tier III	**2.7.3. Tier II**	**2.7.4. Tier I**
ORAT Project Lead may task ORAT resource to support project as needed	ORAT resource(s) may be assigned to manage/support ORAT efforts across multiple projects	Establish dedicated ORAT Team, potentially utilizing external contractors, with responsibilities and workstreams as needed

2.7.5. Step 2: Finalize ORAT Budget

The ORAT Project Lead should finalize the ORAT budget for Project Sponsor approval. The budget should be based on the ORAT Plan and its activities including Familiarization, Training, and Operational Exercises.

For projects where a dedicated ORAT Team is required, budget should be accommodated in the Project's overall

budget. This should have been planned for following the development of the Draft ORAT Plan in the Define Phase of the Project when the resourcing needs become known based on the scale of activities to deliver.

2.7.6. Step 3: Establish ORAT Task Management

Similarly, to delivering any project, the ORAT Project Lead should establish how day to day tasks will be managed with the Project Sponsor. This involves the establishment of review meetings and the utilization of action trackers.

2.7.7. Successful Outcomes:

- ✓ **ORAT Team established** – depending on the Project Tier, the ORAT Team is established to effectively deliver the ORAT activities required by the project.

- ✓ **ORAT Budget established** – the budget needs for ORAT are identified, depending on the initial ORAT Plan.

2.8. Apply Appropriate ORAT Governance & Reporting

Primary Responsibility	ORAT
Key Collaborator(s)	Sponsor
Project Tier Applicability	Varies

This activity refers to the mobilization of the DEN ORAT Governance Framework (Figure 4 below) and application of specific processes and tools to monitor and report that the ORAT Program is on track and delivering its intended benefits. An ORAT Governance Framework is in place at DEN, across all projects, to provide authority and accountability for defining and controlling ORAT activities, outcomes, and ensuring successful go-lives across concurrent DEN projects. At this stage, the project should be established within the existing Governance Framework and decisions made on the appropriate level of application of governance and reporting based on the Project Tier.

Governance is key to the delivery of ORAT as it:

A. Defines clear roles and responsibilities;

B. Provides assurance to key stakeholders and sponsors that the project is on track to deliver against the ORAT Plan and Schedule;

C. Empowers efficient and effective decision making by defining levels of authority;

D. Establishes effective escalation routes for risk and issue management;

E. Establishes decision gates/key milestones for review and approval of planning and execution activities; and

F. Provides confidence to Corporate and Business-As-Usual stakeholders.

At DEN, the overarching ORAT Steering Committee (OSC) oversees all projects. The OSC has full ownership of the business outcome of ORAT Programs and includes senior representatives from the operations groups, key stakeholders, and current Project Managers. The OSC addresses key risks and issues that will impact the likelihood of a successful opening and agrees suitable actions to mitigate them. It can implement agreements and recommendations, as well as the authority to escalate issues to the Executive Leadership. The OSC has developed a Charter, the purpose of it is to establish a partnership between the ORAT Team and key stakeholders. This Charter defines several key elements of the ORAT process and establishes guidelines for decision-making, communication protocols, risk management strategies, and the overall timeline for achieving key milestones.

For each project, an ORAT Management Group should be established that is responsible for the ORAT application per project, and ORAT Working Groups if required (determined by Project Tier).

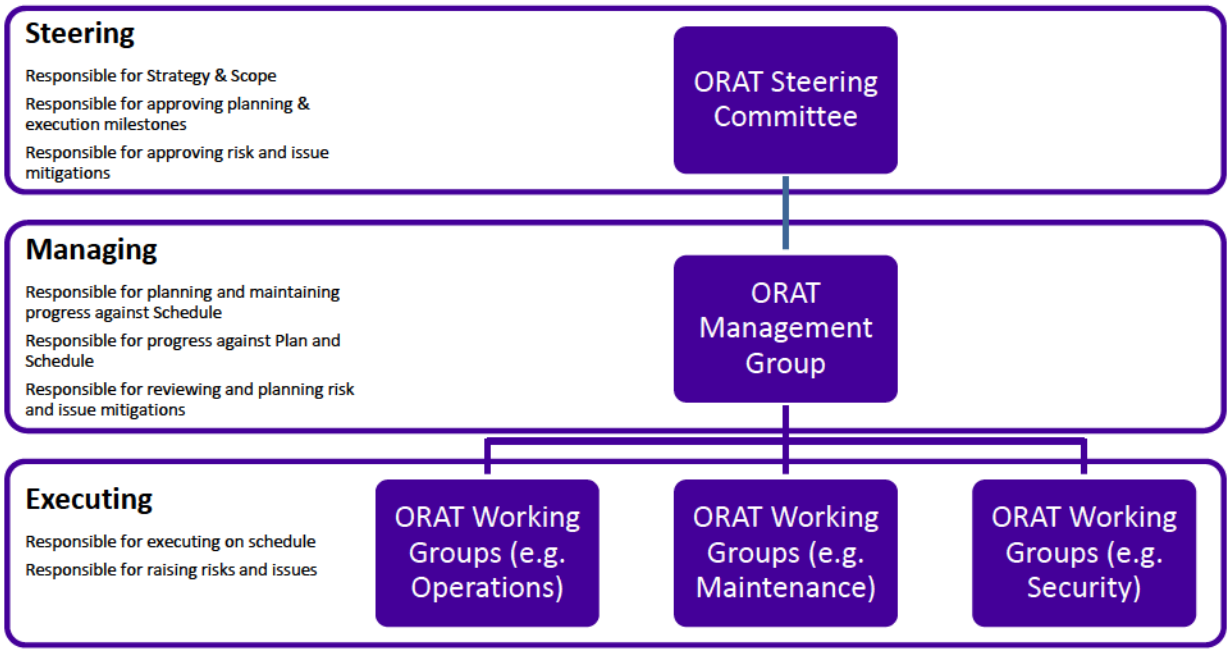


Figure 2-2: DEN ORAT Governance Framework

2.8.1. Step 1: Establish the project within the existing OSC

Prior to project mobilization, the ORAT Project Lead should establish the project within the OSC and ensure the correct representation and availability of information. It is anticipated that if a project is in delivery that the relevant PM will attend the OSC as required. The OSC ensures appropriate information sharing and progress reporting to assure stakeholders that the program is on track. They are also responsible for decision-making including approval to proceed with the planning and execution of activities and agreeing mitigation measures for escalated risks and issues.

2.8.2. Step 2: Establish the ORAT Management Group

The purpose of the ORAT Management Group is to plan and manage the execution of the ORAT activities specified in the ORAT Plan. Each project requires its own Management Group, and the group size is determined by the Project Tier.

The ORAT Management Group is led by the ORAT Project Lead and is composed of representatives from stakeholder groups that are key to planning and delivering readiness activities. The Group should not include representation from stakeholders that are just included in project delivery, but those that are critical to designing and taking on tasks to develop ORAT activities.

2.8.2.1. Application at different Project Tiers:

2.8.3. Tier III	**2.8.4. Tier II**	**2.8.5. Tier I**
Likely no Management Group required	Small Management Group made up of 1-3 stakeholder representatives	Medium Management Group made up of 3-6 stakeholder representatives

2.8.6. Step 3: Establish ORAT Reporting

Ensuring stakeholders are provided regular, transparent progress updates on ORAT is key to good ORAT

governance and effective project delivery. Reports should be sent to key stakeholder leaders with the status of readiness activities, milestones achieved, and any risks and issues. By keeping everyone in the loop with updates on a weekly, monthly, or quarterly basis you help provide accountability and plan any mitigations efficiently. The main aim of the report is to escalate risks and issues.

The ability to generate summarized reports on the progress of the different ORAT activities, relative to an initial baseline plan, will enable key stakeholder leaders to have real time access to information as required. A management system to record progress information and to generate user definable reports will simplify reporting activities and should ensure more accurate and auditable reports are generated. The types of metrics that you report on will vary depending on the stage of the program but typical metrics might include:

A. Number of SOPs to be developed and progress to completion;

B. Number of impacted stakeholder groups and ongoing engagement assessment;

C. Number of technical or operational training materials to be developed and progress to completion;

D. Training Design and Delivery progress against milestones;

E. Exercise Design and Delivery progress against milestones;

F. Number of stakeholders attending training and familiarization activities;

G. Space readiness status;

H. Risks identification, management and mitigation; and

I. Issues identification, management and resolution.

2.8.7. Successful Outcomes

- ✓ **Project established in existing OSC** – the project should be established within the standing OSC.
- ✓ **ORAT Management Group established** – depending on Project Tier, the Management Group should be established by the ORAT Project Lead.
- ✓ **ORAT Reporting established** – cadence and audience of reporting should be agreed utilizing existing reporting tools and templates and aligned to the Stakeholder Engagement and Management Strategy.

2.9. Define Phase Toolkit

Checklist	Templates	Reference Resources
☐ Project Tier ☐ Impacted stakeholder list ☐ Stakeholder Engagement & Management Strategy ☐ Stakeholder Contact List ☐ Operations and Maintenance Budget Forecast ☐ ORAT Plan and Go-live Milestone Schedule ☐ Risk Register ☐ Issues Tracker ☐ ORAT Governance and Reporting	[Project Management Plan (PMP)] [Risk Register] [Issues Tracker] [ORAT Plan] [Go-live Milestone Schedule] [ORAT Progress Report]	[Refer to the Project Management Manual and Project Life Cycle (PLC)] [ORAT Steering Committee Charter] [PLC Project Categorization Methodology]

Chapter 3- Deliver Phase

The Deliver Phase is broken into three subphases – Prepare, Plan, and Execute. The three subphases have distinct activity chevrons but the entire phase is encompassed by a loop of ongoing underlying activities indicated by the orange circle.

The underlying activities include:

A. Ongoing Governance & Reporting;

B. Risk & Issue Management & Task Management; and

C. Ongoing Stakeholder Check-ins & Design Reviews.

Each of these activities is initiated, and the responsibilities and involved parties are established during the Define phase (Section 2.4, Section 2.7, Section 2.9). To ensure alignment between DEN and the project team, maintain timely progress, and preserve the vision and needs of all impacted stakeholders and facilities, regular program management and monitoring are essential. The ORAT Project Lead is responsible for overseeing these activities, and a schedule of meetings and check-ins should be established in the Define Phase and followed consistently throughout the Deliver Phase of the project.

3.0. Prepare

3.0.1. Introduction

Purpose: To start putting some structure around the ORAT Program by developing trackers; reviewing operational impacts and defining system and asset responsibilities.

Considerations:

A. What ORAT resources are needed to successfully complete the project / program?

B. Is this something we can manage inhouse or do we need to look externally?

C. How new, different or changing are the processes and systems of the project / program?

D. How impactful will this be on our stakeholders?

E. How do we ensure all parties are aligned on the future end state?

9.1 Operational Charrette

Primary Responsibility	Designer
Key Collaborator(s)	ORAT
Project Tier Applicability	Varies

An operational charrette is a collaborative meeting during which a range of stakeholder representatives discuss the vision and ideas to inform design options. The intent of the Operational Charrette at DEN Project Mobilization is to involve end-user operations and maintenance groups in the early operational process to ensure their needs are captured in the requirements for the design. Engaged designers should facilitate the session and ensure a robust information capture and documentation process. The ORAT Team will support on-going design reviews and page turns as defined in the relevant DSM.

3.0.2. Step 1: Schedule Operational Charrette

The Operational Charrette should happen early on in project delivery – before detailed design is completed. The ORAT Project Lead should ensure it happens as part of the process and support the designer by inviting relevant project stakeholders, frontline staff, and end-users to participate. The ORAT Project Lead should ensure end-user operators and maintainers are sufficiently represented to provide input.

3.0.3. Step 2: Deliver Operational Charrette

The designer, in close collaboration with the ORAT Project Lead, should facilitate a collaborative Operational Charrette, capturing design priorities, including Operational Requirements. In large projects, several charrettes should be delivered across the relevant functional areas.

Principles for delivering the charrette(s) include:

A. Defining goals that are obtainable and yet demanding, so that expectations can be managed.

B. Building out a schedule that factors in breaks and keeps a demanding pace.

C. Trying to find a location on the project site, which will get everyone off their home turf.

D. Strive for diversity in authority, culture, gender, and expertise, and know the personalities and group dynamics ahead of time, if possible.

E. Participants should feel listened to and considered equally during the sessions.

Example questions to utilize during the charrette(s) to capture end-user operational needs and input include:

A. How are operations conducted today in this area? What systems and tools are used? What supporting infrastructure and spaces are required?

B. What challenges are faced today in this area?

C. What would make operations more efficient and effective? What innovations could be incorporated in future?

D. What DEN Standards need to be adhered to?

E. What are the performance needs that must be met by the operation?

3.0.3.1. Application at different Project Tiers:

3.0.4. Tier III	**3.0.5. Tier II**	**3.0.6. Tier I**
Workshop with Primary	Single operational charrette	Multiple operational charrettes

Stakeholder group may be sufficient	involving multiple stakeholder groups	may be needed to ensure participation from all project stakeholder groups, and key frontline staff and end-users

3.0.7. Step 3: Develop Operational Requirements

The designer should produce a summary of initial Operational Requirements for the design in a clear and concise document. The Operational Requirements should detail how the product will be used to satisfy a user's need and the necessary components to ensure it optimizes operations and is future proofed.

3.0.8. Step 4: Prioritize and Select Operational Requirements

An exercise should take place to ensure that the outputs from the Operational Charrettes are reviewed and prioritized. The priority list should not become a list of nice-to-haves but rather a considered list of implementable requirements for design development.

This review and prioritization exercise should be undertaken by the Project Manager and Project Sponsor.

3.0.9. Successful Outcomes:

- ✓ **Operational Requirements for the design captured and prioritized** – through a facilitated process led by the designer, requirements should be captured that acknowledge how the product will be operated and maintained.

- ✓ **End-User groups included in early design stage** – involving end-user groups, such as operators and maintainers, early in the design process means the design is more likely to meet their needs and avoid costly changes later or inefficiencies.

3.1. Develop Detailed ORAT Plan & Activation Requirements

Primary Responsibility	ORAT
Key Collaborator(s)	PM
Project Tier Applicability	All

This activity focuses on the further development of the ORAT Plan in the PMP and defining Activation Requirements to support and align ORAT decision-making. These are measurable standards stakeholders can use to determine if the project is ready for the transition from construction to operations. The following approach will be used to develop the detailed ORAT Plan and establish Activation Requirements for the project.

3.1.1. Step 1: Develop Detailed ORAT Plan

At roughly 80% design, the ORAT Team should finalize the ORAT Plan that was initially developed in the Define Phase by detailing the type and target number of activities required. While the initial ORAT Plan in the Define Phase considered what activities may need to be delivered, this ORAT Plan provides confirmation on what exactly will be delivered, when and how. For example, the purpose, type and number of training, familiarization and operational exercises should be clearly planned with the materials required to support them and intended timeframes. The Detailed ORAT Plan should consider the following components:

A. ORAT Governance including risk management and reporting processes
   a. What is the agreed governance structure for the project?
   b. How are risks being managed on the project?
   c. How is ORAT progress reported and to who?

B. Program Management including team roles and task management
   a. What team roles and responsibilities are in place?
   b. How is the team being managed, and tasks allocated?

C. Construction and Commissioning Alignment
   a. What interfaces are in place with construction and commissioning activities, if any? What dependencies are there on construction and commissioning activities?
   b. How will construction and commissioning teams be engaged with to ensure the right information is received at the right time? Who are the key contact groups?

D. Operations Planning
   a. What operations and maintenance processes need to be reviewed and with which groups?
   b. How will these reviews be conducted?
   c. What plans and documents need updating?
   d. When will this be done?

E. Training & Familiarization
   a. What training and familiarization is required? For which stakeholders?
   b. What contractual training will be delivered, if any?

F. Operational Exercises
   a. What Operational Exercises are required? What type (e.g., Drill, Functional, Full-scale or Emergency)?
   b. When will these Operational Exercises take place?
   c. Who are the key stakeholders?
   d. What feedback loop is there to update process documents?

G. Transition & Go-Live
   a. How will project Go-Live be phased? How will cutover occur?

b. Which stakeholder groups are involved in the transition?
c. When will the moves take place?

The Detailed ORAT Plan will be a living document throughout the Program and can be updated if required.

3.1.2. Step 2: Develop Activation Requirements

Forming a collective vision and agreeing expectations for Activation is a crucial part of the Deliver Phase: Prepare. The ORAT Project Lead should facilitate leaders from key stakeholder groups and the OSC members in a 'Vision for Opening' workshop to articulate and agree Activation Requirements that define a shared vision for what should happen on Opening Day. These Requirements guide future ORAT decision-making.

The Vision for Opening Workshop should serve to meet the following objectives:

- align stakeholders on a common vision for Opening Day;
- define the exact Activation Requirements that must be in place for Opening to occur; and
- prioritize any activities if required.

These Activation Requirements should specify the minimum thresholds for the safe operation of the airport and for meeting business objectives. Early identification of these minimum requirements is essential as they guide the prioritization of ORAT activities and the OSC will use the Activation Requirements to feed into the Transition and Cutover Planning and support decision making as to whether the transition to Opening Day can occur.

3.1.3. Successful Outcomes:

- ✓ **ORAT Plan updated in PMP** – the ORAT Plan template is updated to include exactly what will be delivered for the project, as well as, when and how.
- ✓ **Go-Live Milestone Schedule updated** – the full list of planned activities defined in the updated ORAT Plan are visualized in a Schedule to provide continued awareness of ORAT Needs.

3.2. Review Operational Impacts

Primary Responsibility	ORAT
Key Collaborator(s)	PM
Project Tier Applicability	Varies

This activity focuses on reviewing operational impacts by conducting an Operational Impact Assessment. The outcome of the Operational Impact Assessment is a document that provides a high-level overview of the processes, people, and systems impacted by the project. It enables stakeholders to understand the scope of the project change and its implications to their operations. Ultimately, its aim is to identify what the ORAT Program needs to address through activities such as process updates, training and familiarization, and operational exercises.

3.2.1. Step 1: Conduct Operational Impact Assessment Workshop(s)

The Operational Impact Assessment is produced through a series of workshops, facilitated by the ORAT Team, led by the ORAT Project Lead, that systematically evaluate the final design against current operations to assess the project's impact on people, processes, and systems. Attendees should include representatives from impacted stakeholder groups including frontline staff who know the operation well.

Depending on the Project Tier, several workshops may be required to facilitate effective sessions for stakeholders. To aid delivery, operations can be organized into operational flows and steps, and workshops could be delivered to cover relevant flows and steps depending on the session length. Operational flows for an airport include passengers, baggage, aircraft, staff, goods, and waste. Each operational flow can be further divided into a series of logical steps in sequence. For example, the passenger operational flow might consist of the following steps: arriving at departures, checking in, security screening, pre-boarding, boarding, deplaning, baggage claim, and exit to ground transportation. For each operational step that the project impacts, impacts of the new design should be assessed utilizing the following criteria:

New People – does a new role need to be created, or an existing job description updated? For example, adding a larger terminal footprint will impact janitorial contracts.

New Process - does an existing process need to be changed, or is a new process required? For example, introducing automated bag drop units at check-in will affect check-in processes. This is a change for the operation that should be trained and tested prior to go-live.

New System – is a new technology or asset being introduced? A new system being implemented will have impacts on wider integration, and the people and processes utilizing it. Therefore, it should be tested and staff prepared to utilize the system.

Typically, the first portion of the workshop should provide an overview of the new design as attendees are likely unfamiliar. This will require attendance from someone from the design team who can present it in detail. Once the design has been presented, attendees should discuss and agree the impact on people, process, and systems for each operational step as high, medium, or low.

3.2.1.1. Application at different Project Tiers:

3.2.2. Tier III	**3.2.3. Tier II**	**3.2.4. Tier I**
1-3 Workshops may be required to complete the Operational Impact Assessment with a small number of stakeholder groups represented	4-8 Workshops may be required to complete the Operational Impact Assessment with a medium number of stakeholder groups represented	9-20 Workshops may be required to complete the Operational Impact Assessment with a large number of stakeholder groups represented

Step 2: Document Operational Impact Assessment

The Operational Impact Assessment document should consolidate the agreed high, medium, or low impact for each operational step. This then informs operations planning and readiness activities planning as the focus should be on high and medium impacts.

3.2.5. Successful Outcomes:

- ✓ **Complete Operational Impact Assessment** – the Operational Impact Assessment workshops are completed with target stakeholders, operational impacts are documented, prioritized and agreed with stakeholders to begin operations planning and readiness activities planning.

3.3. Define System & Asset Responsibilities

This activity brings together technical, operational, and delivery team stakeholders to define system and asset responsibilities. This ensures there is clarity on roles regarding new systems and assets (e.g. who is maintaining it) and allows for subsequent readiness activities to be planned for appropriately.

Primary Responsibility	ORAT
Key Collaborator(s)	Mx, AM
Project Tier Applicability	All

3.3.1. Step 1: Develop Owner, Operator, Maintainer, User (OOMU) Responsibilities Matrix

The ORAT Team should develop an Owner, Operator, Maintainer, User (OOMU) Responsibilities Matrix document for each system or asset type during all project phases and the end state. The following approach should be used:

A. Acquire existing Responsibilities Matrix – is there an existing document? If so, this document should be used as the baseline for updates.

B. Focus on scope of all impacted systems – what systems are new or impacted by the project? Utilizing the Operational Impact Assessment, only focus on the systems impacted by the project and those it integrates with.

C. Understand the system specifications – are there any contractual requirements that impact responsibilities of the new system? Review contract documents to understand any requirements.

D. Align stakeholder groups - stakeholders will likely need facilitated discussions on who will inherit the new systems and to agree what responsibilities (Owner, Operator, Maintainer, or User) will be.

E. Document a new or updated Responsibilities Matrix - ensure a clear and concise document is created with version control and is shared with relevant stakeholders.

The Matrix should be reviewed throughout a project and beyond, and any changes should be communicated to stakeholders.

3.3.2. Successful Outcomes:

- ✓ **System & Asset responsibilities agreed and documented in the OOMU** – the OOMU documents agreed stakeholder responsibilities for each system or asset.
- ✓ **Periodic OOMU reviews** – any changes in system or asset responsibilities are captured in the OOMU and communicated to stakeholders.

3.4. Develop Detailed Operations & Maintenance and Asset Management Plans

Primary Responsibility	Ops, Mx, AM
Key Collaborator(s)	ORAT
Project Tier Applicability	All

Once impacts have been understood and responsibilities for systems defined, detailed plans for how operations, maintenance and asset management will be conducted should be developed. These Operations & Maintenance and Asset Management Plans should be updated by the relevant groups, facilitated by the ORAT Team.

This section refers to the development of DEN Operations and Maintenance Plan documents, rather than contractually provided O&M Manuals as part of a construction handover process. The following approach should be used to ensure airport stakeholders and the ORAT Team are aligned in their development of the O&M and asset management plans.

3.4.1. Step 1: Develop Draft Operations and Maintenance Plans and Asset Management Plans

At this stage, an intentional document review should be conducted by Operations and Maintenance teams to update any Operations and Maintenance Plans, such as the Preventative Maintenance Plan, and by Asset Management teams to update Asset Management Plans to begin preparations to take over ongoing operation and maintenance responsibilities and manage assets.

Information, such as asset data and warranty information, provided by the Contractor or Supplier is critical to undertaking these updates. If draft O&M Manuals or asset information has not been received at this point in the project, then Specifications can be referred to. Later, the Plans can be updated as formal documents are received and install begins (e.g. asset nameplate information becomes available).

The ORAT Team should facilitate the establishment of expectations on what information is included in the documentation issued by the Contractor (e.g., asset warranty information, commissioning documentation, O&M manuals, preventive maintenance schedules, asset data, etc.) and support the maintenance, operations, asset management and facilities management groups to update their Plans and work with the Contractor to review draft O&Ms, spare parts, and warranty information.

3.4.2. Step 2: Update Service Contracts

Due to the specialized nature of certain building systems and/or to supplement airport staffing, DEN may outsource operations and maintenance services for select systems. To prepare for the day-to-day operations and maintenance of new facilities, airport stakeholders may need to establish new or modify existing service contracts as well as review additional staffing requirements to support the additional equipment and building footprint. Operations and maintenance groups should update their Service Contracts, supported by the ORAT Team to facilitate providing reference information.

3.4.3. Successful Outcomes:

- ✓ **Draft Operations and Maintenance and Asset Management Plans developed** – existing Plans should be reviewed and updated using new system information referencing any available contractor/supplier-provided asset information at this stage in the project.

- ✓ **Service Contracts updated** – to prepare for operations and maintenance, any service contracts should be updated to ensure seamless asset integration.

3.5. Deliver Phase: Prepare Toolkit

Checklist	Templates	Reference Resources
☐ Operational Design Charrettes ☐ Operational Requirements ☐ ORAT Staffing & Budget ☐ Detailed ORAT Plan ☐ Activation Requirements ☐ Operational Impact Assessment ☐ OOMU ☐ Draft Operations and Maintenance Plans ☐ Asset Management Plans ☐ Service Contracts	[Operational Impact Assessment] [OOMU] [Operations and Maintenance Plans] [Asset Management Plans]	[Refer to the Project Management Manual and Project Life Cycle (PLC)] [Project Management Plan (PMP)] [Project Delivery/Construction Schedule]

End of Chapter

Chapter 4- Deliver Phase

4.0. Plan

4.0.1. Introduction

Purpose: To define new or updated processes; identify people readiness needs along with designing Training and Familiarization materials and Operational Exercises and develop a Transition & Cutover plan to all feed into an Integrated Schedule.

Considerations:

A. Can we ensure we are defining only new or changing processes?

B. Can we use operational exercise planning to engage contractors and align expectations?

C. How do we ensure stakeholder ownership of Operational Exercises?

D. How do we ensure we are meeting stakeholder training needs?

E. How do we ensure Commissioning inform and are aligned with the Transition and Cutover plans?

F. How do we ensure enough time is factored into the Master Schedule for all the ORAT activities?

4.1. Identify & Develop SOPs and Contingency Plans

Primary Responsibility	Ops
Key Collaborator(s)	ORAT
Project Tier Applicability	All

As a result of the scope of change of the project (e.g. a new system or new facility footprint), new operational processes may need to be defined or updated. The changes required should have been identified in the Operational Impact Assessment. The following approach should be used to ensure impacted stakeholders, identified in Section 2.4 above, are engaged to define SOPs; Work Instructions and Contingency Plans with ORAT support as needed.

4.1.1. Step 1: Identify SOPs and Contingency Plans

After completion of the Operational Impact Assessment, each stakeholder group will have awareness of processes needing to be developed or updated at a high-level. Further analysis should be conducted to:

A. Identify any operations SOP documents or procedural manuals that need updating;

B. Identify any Contingency or Emergency Plans that need updating; and

C. Identify any further overarching/high-level processes that may not be formal documents but need to be updated and communicated to stakeholders.

4.1.2. Step 2: Develop or Update SOPs

Processes identified as having a high or medium level of impact associated with them as they are new or changing should be intentionally discussed and developed/updated with key stakeholders.

The ORAT Project Lead will can either:

A. Facilitate stakeholder workshops to identify, discuss and agree new or changing processes with support from experts to provide any required technical information; or

B. Monitor stakeholders who wish to discuss internally.

In facilitated workshops, it is important to hear from staff conducting these processes to ensure that every aspect is understood and planned for in the new facility. For complex processes, Visio Process Maps, or similar format, can be used to align stakeholders. It is also important in workshops to ensure a focus on the 'future-state' process as opposed to the 'current-state' process. Once new or updated processes have been agreed, it is the responsibility of the stakeholders to document them in the relevant SOP, Work Instruction, or Contingency/Emergency Plan.

Additionally, process validation exercises can take place at any point after the process has been documented when it is beneficial to align stakeholders and confirm understanding. It is important to have validated the process through a tabletop discussion before the delivery of an Operational Exercise within the new facility.

4.1.3. Successful Outcomes:

✓ **Complete process review** – each stakeholder group confirms processes have been reviewed against the Operational Impact Assessment and that relevant SOPs and Contingency / Emergency Plans are up to date.

4.2. Determine People Needs & Prepare Training & Familiarization Materials

The new facility relies upon people to function and staff preparedness is essential for the operational success of the facility on Opening Day. Comprehensive training programs should be implemented to ensure that end users, maintainers, and operators are completely familiar with the terminal by Opening Day, and technically trained to utilize the new facility and its assets for their job role. The following approach should be used to ensure training needs are defined and training materials prepared.

Primary Responsibility	ORAT
Key Collaborator(s)	PM
Project Tier Applicability	All

4.2.1. Step 1: Conduct Training Needs Analysis

A Training Needs Analysis (TNA) provides the opportunity to have a consistent and reliable way of capturing stakeholders' training needs. The TNA should be conducted by the ORAT Project Lead through stakeholder discussions and in consultation with the Project Team.

The Operational Impact Assessment, updated SOPs, and draft Operational & Maintenance Plans and Asset Management Plans should inform the TNA. The TNA forms the baseline of the people readiness program and detailed planning of familiarization activities, technical training delivery, and any recruitment and induction requirements. The TNA should identify:

A. Training needs due to changing processes and systems;

B. Training level required depending on job role; and

C. Familiarization requirements because of new systems/facility/processes.

4.2.2. Step 2: Develop Training & Familiarization Schedule

The completed TNA should provide a detailed analysis of what training is required and to what level. This should be used to refine the Training & Familiarization Plan within the Detailed ORAT Plan into a Training & Familiarization Schedule with the specific activities that will be conducted and when they will be delivered. The Schedule should also identify which trainings may require annual renewals/reviews or to be used in future onboarding. Ensure Training & Familiarization sessions are scheduled as early as possible, and that attendance is tracked and verified. Types of Training & Familiarizations include:

Table 4-1: Types of Training & Familiarizations

Type	Description	Examples
General Facility Familiarization	Provides an overview of the general facility, spaces, and functions and covers aspects that stakeholders will encounter during their day-to-day activities.	Staff entry points, paths of travel, location of amenities, employee parking, vendor delivery areas, waste disposal, use of building technology.
Public Safety and Security Familiarization	Orientation in life safety systems, emergency, and security procedures.	Evacuation routes, assembly points, access for first responders, location of emergency equipment and phones.
Systems Familiarization	Orientation in basic operation and procedures of airport systems to enhance contractual training.	Shared use terminal equipment, baggage handling equipment, gate operations, exit lane equipment.

Operational Training	Stakeholder-specific training focused on the areas, systems, and functions relevant to the stakeholder being trained. This training may cover all the systems and operational processes.	Handling of oversize inbound and outbound baggage, swing gate operation, passenger processing associated with arriving and departing passengers, aircraft pushback procedures.
Technical Training (see below)	Supplier-led training on newly installed and commissioned systems.	Air Handling Units, Glycol Mixing pumps, advanced lighting controls, emergency communication, conveyance systems, energy management components, systems connected to piping and cuts, access hatches.

4.2.3. Step 3: Support Technical Training

If new systems are being implemented by the project, it is likely that the manufacturer or supplier will have a contractual obligation to deliver Technical Training. Technical Training is typically focused on system operation and maintenance and is geared towards maintenance or technical personnel. It has specific requirements defined in the contract such as number of participants and number of sessions.

Coordination of the required training with suppliers can be integrated into the ORAT Program. The ORAT Team can review training curriculums to ensure they meet the end-user's needs and track and assure that quality delivery takes place for all systems across the project. This should be done in close collaboration with the Project Manager and Project Engineers to ensure there is not an overlap of responsibility. Useful tools for ensuring end-user needs are met include:

A. Training Plan Review Checklist – to ensure it meets training best practices such as clear learning objectives, knowledge checks, and accessible content.

B. Session Sign-In Sheet – to enable effective tracking of attendance.

C. Session Feedback Form – to assess that attendees feel confident they can now operate or maintain the system.

Note, beyond technical training, stakeholder participation in the wider commissioning process, such as witness testing, should be encouraged as it provides opportunity for further learning and familiarity.

4.2.4. Step 4: Prepare Training & Familiarization Materials

Familiarization is normally generic and the responsibility of the ORAT Team. Familiarizations are made to provide awareness of facility spaces, systems, operations, and/or processes. It has very few limitations and can be adjusted or presented as many times as required to accomplish the operational objectives. Familiarization programs are typically delivered on a train-the-trainer basis, and are supported by maps, booklets, breakroom videos, reference cards, and other collateral created for the purpose of educating stakeholders on the spatial changes being made to the Airport as part of the project. A familiarization journey allows for customization of materials such that individual learning styles, preferences, and circumstances can be taken into consideration when choosing a communication method.

4.2.5. Successful Outcomes:

- ✓ **Training Needs Analysis completed** – consolidated list of all impacted stakeholder training and familiarization needs.
- ✓ **Training & Familiarization Schedule** – sessions are schedule as early as possible.
- ✓ **Technical Training Support** – training curriculums reviewed to ensure user needs will be met and quality delivery takes place.

4.3. Agree & Design Operational Exercises

Delivery of a new or modified space/system/facility should always undergo a thorough Operational Exercise Program to ensure it meets end-user requirements for operation. Operational Exercises are a series of stakeholder-focused tests that verify staff readiness to operate systems and processes, under different operating states (normal, irregular, and emergency operations). The Operational Exercise Program should be designed to test the areas of greatest risk - new and changing processes, and any new systems and facility spaces as indicated by the Operational Impact Assessment.

The Program should be progressive, starting with singular systems and processes in Tabletop exercise and Drills and ultimately leading to a Full-scale operational exercise featuring volunteer passengers if required by the Project Tier. Operational Exercises vary in size, complexity and overall execution and are adapted to closely simulate actual operating conditions. Contingency processes and emergency situations should also be tested to ensure staff and facility readiness. It is critical to capture and address any operational issues that are identified during an Operational Exercise to ensure minimal impact during live operations. The following approach will be used to ensure Operational Exercises are identified and designed appropriately.

Table 4-1: Types of Operational Exercises

Type of Exercise	Description	Examples
Tabletop	Single or multiple process review before the ability to physically test and verify.	Review of vehicle service routes.
Drill	Single process testing (discrete processes, singular function, asset of isolated process), typically under normal operating conditions.	Landside check-in; Wayfinding, Aircraft fit checks.
Functional	Multiple process testing (more than one linked process, multiple assets, requires volunteers).	Outbound passenger flow through check-in and security.
Full-scale	Multiple process and high-volume pressure testing (linkages, integration of processes). Operation of combined systems, end-to-end processes, and integrated operational scenarios. Also provides an opportunity to validate corrected issues from previous Drills.	Integration of inbound and outbound baggage handling with departing, arriving, and transfer passenger processing.
Emergency	Tests of emergency operation, command and control, and communication procedures. Emergency Exercises may include a series of orientation and familiarization exercises for first responders, and for personnel involved with emergency operations and procedures.	Terminal evacuation; Active shooter training; Explosive device disposal.

4.3.1. Step 1: Identify Exercise Needs

The ORAT Team should identify the Operational Exercises that are required as a function of new or changing processes and systems, informed by the Operational Impact Assessment. A Tabletop and/or a Drill should be identified for each new or changing process at a minimum. Functional, Full-scale, and Emergency Exercises should be delivered if the project has a larger footprint with integrations across many stakeholder processes and systems.

The ORAT Team should develop the list of required Operational Exercises and seek approval from the OSC. Depending on the Project Tier and the number and type of exercises needed, the ORAT Team should begin

coordination with the DEN Training & Exercise Team at this Step to ensure alignment and any required support. This will likely only be required for Tier 2 and 3.

4.3.1.1. Application at different Project Tiers:

4.3.2. Tier III	**4.3.3. Tier II**	**4.3.4. Tier I**
Functional and Full-scale Exercises are unlikely to be used at this Project Tier	Full-scale Exercises are unlikely to be used at this Project Tier	Full-scale Exercises may be required at this Project Tier

4.3.5. Step 2: Design Operational Exercises

Once the list of required Operational Exercises has been agreed with the OSC, exercises will be designed to simulate real operations. Clear objectives for each Operational Exercise must be agreed and evaluation criteria determined. Scenarios will be designed to prove the exercise objectives and scripts developed for the different roles in each scenario. Making the Operational Exercise as "real" as possible will benefit both the staff and the exercise results.

For each agreed Operational Exercise, the ORAT Team will collaborate with process/system/space owners, subject matter experts (SMEs) and end-users to design an Operational Exercise Plan defining the following items:

A. Owner - who has accountability for its delivery and its pass/fail decision.

B. Objectives - what the exercise is proving.

C. Exercise Evaluation Guide- what constitutes a success vs a failure (e.g., KPI achieved, quantitative & qualitative measures attained).

D. Scenarios - what scenarios will be delivered to meet the exercise objectives.

E. Location & Footprint - where the exercise will take place.

F. Dependencies - what systems need to be functional, what processes need to be agreed, what approvals granted and what training delivered prior to execution.

The Operational Exercise Plans will be presented to and reviewed by the OSC for final approval before Operational Exercises are delivered in the Execute Phase. Functional and Full-scale operational exercises may require the use of volunteers, which may incur costs requiring approval.

4.3.6. Step 3: Schedule Operational Exercises

The ORAT Team should schedule preliminary Operational Exercise dates, in coordination with end-user stakeholders referencing latest activation dates. Operational Exercises require unimpeded access to building areas and functioning systems and are therefore scheduled in close collaboration with the Project Team.

Tabletop exercises can take place any time. Drills should take place as soon as systems are ready, and the space can be cleared for access. Drills can typically be delivered in a Personal Protective Equipment (PPE)-environment (please ensure you link in with the PM to ensure compliance based on the specific work areas). Larger exercises such as Functional, Full-scale, and Emergency likely require the space to be PPE-free and therefore are typically timed to start after the Temporary Certificate of Occupancy (TCO) construction milestone.

Time for coordination, pre-operational exercise inspections, ensuring safety, readiness, etc. and then Go/No-Go

decisions should be built into the Schedule. Additionally, consider allocating time for corrective action, re-running operational exercises, and rescheduling.

4.3.7. Successful Outcomes:

- ✓ **Identified and designed exercises** – exercises are identified and designed through the development of Operational Exercise Plans.
- ✓ **Scheduled exercises** – a timeline of which Operational Exercises will take place when is established.

4.4. Develop Transition and Cutover Plan with Commissioning alignment

The objective of the transition & cutover planning activity is to plan how the transition and cutover will take place – what time it will occur, if it will be phased, and what primary activities are needed to support the transition. The following guiding principles should be considered:

Primary Responsibility	ORAT
Key Collaborator(s)	PM
Project Tier Applicability	All

A. Opening Day should be driven by a state of readiness, rather than a date;

B. Phased Opening can mitigate impacts to stakeholders;

C. Sufficient time should be given post TCO, pre-Opening Day for final fit-outs; Operational Exercises and Issue Resolution;

D. Commissioning activities should be completed prior to cutover; and

E. No hard opening dates should be published in the media before there is assurance that these dates can be met.

4.4.1. Step 1: Develop Transition and Cutover Plan

The ORAT Team should develop a Transition and Cutover Plan informed by the Activation Requirements defined in the Deliver Phase: Prepare. These Plans are crucial for ensuring that all necessary elements are in place for the transition and cutover, facilitating the smooth movement of people and equipment into the space, and the commencement of operations. Effective logistical planning will help mitigate risks during Activation. This Plan should also consider:

A. Opening Strategy - there are a range of Opening Strategy and Transition tools available, and one should be selected as per the nature and risk profile of the project. All opening strategies must be carefully coordinated with end-users and stakeholders to minimize service interruptions and/or reductions in capacity. Opening Strategies include:

Table 4-2: Types of Opening Strategies

Type of Opening	Description
Phased	Progressive opening of phases, a ramp up into full operation
Big Bang	One large scale opening with no ramp up
Hard Opening Date	Follows a strict deadline that must be achieved and cannot slip.
Soft Opening Date	Follows a proposed date that allows for slippage should more time be needed to revisit project requirements.

B. Move Plan - it is important to identify requirements around the physical move component early in the project. Moves are widely recognized to be stressful, expensive and time-consuming and as such it is important to engage early on with a Move Coordinator (if applicable) to support the planning and execution to ensure a seamless move. Aspects to be considered are:

   a. type of move (e.g. Airport, project, or tenant led);
   b. roles and responsibilities;
   c. logistics requirements;
   d. schedule dates for the moves - consolidated, phased or overnight moves; any opportunities for early moves;

e. inventory of items to be relocated including furniture, equipment, services, points of connection, and IT requirements;
f. operational dependencies (e.g., Business Technology & IT, Systems and Facilities, Staff and Goods, Routes in/out, and Emergency Access & Evacuation);
g. IT coordination for common-use equipment and interfaces with airline/tenant host servers;
h. any specific requirements or proprietary processes that might not be supported by the Move Coordinator (if applicable) and may require additional coordination such as interfaces to airlines/tenant technology;
i. decommissioning of spaces, equipment, or services that are being vacated;
j. construction coordination and alignment; and
k. cutover contingency / fallback planning.

4.4.2. Step 2: Develop Transition Checklists and Review

In collaboration with Commissioning, and DEN stakeholders, the ORAT Team will develop a series of Transition Checklists to track the state of readiness moving towards Opening Day. The Transition Checklists are developed to document and track 90/60/30-day milestones to Opening Day. And the Room Readiness Checklists are developed to assure the readiness of each room for each tenant.

A primary component to the Checklist development includes inventorying all transition activities that are required to be completed by the Airport, project team, agencies, and tenants. Checklists are used for tracking, reporting and communicating status of transition activities. Transition Checklists are comprehensive lists of tasks, related to either a system, process or facility, that need to be accomplished to support the opening. Unlike construction or commissioning checklists, Transition checklists are stakeholder focused and tasks are grouped by working group. The Transition Checklist development is a collaborative effort, led by the ORAT Team, and assisted by stakeholders, tenants, agencies and contractors to validate. After the Checklists are finalized and approved by stakeholders, the ORAT Team manages them, reviews progress relative to the construction schedule and project milestones, updates them as necessary, and reports back on status. The completed Transition Checklists may be employed to confirm stakeholder sign-off. The level of detail and type of checklist required for a project largely depends on the size and scope of the project.

4.4.3. Successful Outcomes:

- ✓ **Minimal risks to Activation** – Transition and Cutover plans sufficiently reduce risks during activation through considered logistical planning and scheduling of transition activities.

4.5. Develop Integrated Schedule

Primary Responsibility	ORAT
Key Collaborator(s)	PM
Project Tier Applicability	All

A Program Schedule should be developed, or coordinated, by the Project Manager that integrates construction, commissioning, and ORAT activities for the project. For the ORAT Program, this enables the Project Team to be aware of when ORAT activities will take place and their dependencies as well as ensures ORAT is incorporated and woven in throughout the lifecycle of the project instead of a closeout activity. Alignment with the ongoing design of operational exercises and construction activities, facilitates the development of operational readiness countdowns, tracking and reporting of ORAT milestones relative to construction progress, and identification of risks owing to construction issues or delays.

4.5.1. Step 1: Develop ORAT Schedule

The ORAT Team should develop a detailed ORAT Schedule, aligning to a Go-Live Milestone Schedule and incorporating detailed ORAT activities with durations and dependencies. At this point in the project, the specific ORAT activities that will be delivered should be known through the development of a detailed Training & Familiarization Schedule and Operational Exercise planning.

These activities should be established in a Schedule program (e.g., MS Project, MS Excel, P6 or Smartsheet).

4.5.2. Step 2: Align ORAT Schedule in Program Schedule

Using the ORAT Schedule, the Project Manager should ensure ORAT activities are captured in the Program Schedule as an Integrated Schedule. Not all activities need to be included, just those on site and requiring alignment to construction and commissioning milestones.

4.5.3. Successful Outcomes:

- ✓ **Consolidated schedule –** single Go-Live Schedule updated to include construction, commission, and ORAT activities to act as a single source of truth for scheduling of activities.

4.6. Deliver Phase: Plan Toolkit

Checklist	Templates	Reference Resources
☐ New or updated SOPs ☐ Contingency Plans ☐ Training Needs Analysis ☐ Training and Familiarization Plan ☐ Training and Familiarization Materials ☐ Operational Exercise Plan ☐ Design Operational Exercises ☐ Transition & Cutover Plan ☐ Transition Checklists ☐ Integrated Schedule	[Training Needs Analysis] [ORAT Operational Exercise Plan template] [Transition Checklists]	[Refer to the Project Management Manual and Project Life Cycle (PLC)]

End of Chapter

Chapter 5- Deliver Phase

5.0. Execute

5.0.1. Introduction

Purpose: To deliver the Training and Familiarization and Operational Exercises program including System and Facility Activation and Handover.

Considerations:

A. How do we ensure the Training and Familiarization Program is effective?

B. How do we successfully coordinate all the moving parts – System and Facility readiness with people readiness and Operational Exercise delivery?

C. Are there opportunities to support the transition of ownership through operator involvement in testing and commissioning activities?

D. How do we ensure that Opening Day is based on operational readiness not construction completion?

E. How do we ensure a smooth, well-coordinated Transition?

F. Where do we want to be 90 / 60 / 30 days out from Opening Day?

G. When do we publicize the Opening Day in the media?

5.1. Track Asset & System Activation and Handover

At Handover, the formal transition of ownership of systems and facilities from the contractor to DEN takes place. On some projects, assets and systems may be activated into operational use prior to formal Handover at substantial completion but maintenance responsibilities may remain with the contractor/construction team until Handover is complete with all necessary documentation. The following approach should be used to ensure construction and Airport stakeholders track facility and system status throughout the delivery phase to prepare for Handover.

Primary Responsibility	AM, Mx
Key Collaborator(s)	PM, ORAT
Project Tier Applicability	All

5.1.1. Step 1: Develop the Handover Plan/Schedule for Assets and Systems

The Project Manager should provide clear information on when assets and systems will be turned over to DEN, and when DEN assumes responsibility for them. This should be documented in a Handover Plan or Schedule, developed by the Project Manager. Key aspects to consider as part of this Handover Plan include but are not limited to:

A. Document and Knowledge Transfer: O&Ms; as-built drawings; training materials; Service Level Agreements

B. Commissioning and Testing confirmation: Functional testing; acceptance testing and commissioning documentation

C. Asset Lifecycle information: Asset register; maintenance plans; warranty information and spare parts

The ORAT Team should ensure that Asset Management and Maintenance groups are aware of the Handover Plan to enable them to prepare accordingly. The ORAT Team should also review that the Handover Plan aligns with the Activation Requirements regarding which assets and systems are required for a smooth transition to opening.

5.1.2. Step 2: Track Activation and Handover

The ORAT Team should compile an Asset and System Activation Checklist using information provided by the Project Team. The Asset and System Activation Checklist should list all the assets and systems involved in Handover and any dependencies between them (e.g. network rooms).

Onsite inspections need to be undertaken to ensure construction and commissioning activities are proceeding in alignment with the agreed Schedule. This activity is intended to inform ORAT planning and operational risk mitigation, not become an assessment of construction status. The Asset and System Activation Checklist should be used by the ORAT Team to prepare operations and maintenance requirements at the time of transition.

5.1.3. Successful Outcomes:

- ✓ **Agreed requirements and plan for Handover** – all parties agree that the physical asset is complete, meets design criteria, and is "ready for operation".
- ✓ **Complete list of readiness activities** – to easily track, report, and communicate the status of Activation activities with stakeholders.

5.2. Collect Asset Information, Critical Spares & Prepare Maintenance Plans

As the project starts to transition into operations, the contractor or supplier should provide the required asset information and any critical spares to DEN, as required. The Project Manager is responsible for ensuring this information is provided to Asset Management and Maintenance groups to enable them to update asset records and to prepare Maintenance Plans.

Primary Responsibility	AM, ORAT
Key Collaborator(s)	Ops, Mx
Project Tier Applicability	All

5.2.1. Step 1: Collect Asset Information

It is essential that asset information handover is aligned with the project turnover or when the Airport formally assumes responsibility of ownership, including operation and maintenance. Airport stakeholders responsible for maintaining the facility should be notified of this assumed responsibility of ownership, including partial utilization of building systems or areas that may be put into service prior to a building stage completion. The asset information provides the necessary tools for efficient operation and maintenance of the new facility and includes, but is not limited to:

A. Operation and maintenance manuals;

B. Asset data;

C. Warranties (including start and end dates);

D. As-built and record drawings;

E. Attic stock list / critical spares list; and

F. Training material (e.g., record of attendance, training videos, training syllabus).

All this information must be transferred far in advance of Handover to enable the Airport to make decisions and execute other readiness tasks.

5.2.2. Step 2: Acquire Critical Spares List

The critical spares list is an essential component of the asset information that should be provided by the contractor or supplier at Handover. If spares themselves are not received as part of Handover, at minimum, a critical spares list should be provided as identified by the Specifications and Contracts. The PM team should review this list and make the necessary arrangements to ensure spares can be procured quickly in a failure and that operational contracts include future provision of critical spares. The ORAT Team should track that this activity takes place and support the PM team to receive information from the Project Manager, if needed.

5.2.3. Step 3: Prepare Maintenance Plans

Earlier in the project, Draft Maintenance Plans should have been developed by Maintenance teams using available information and approved by Asset Management for the first year. At this stage of the project, further refinement and finalization of Maintenance Plans should take place. If Draft O&M Manuals have been received by the Project Team at this stage, Maintenance teams can reference them to update DEN Maintenance Plans for the new facility and/or systems before the final O&M submittal is received.

To ensure a smooth transition and rapid generation of benefits from the investment, this information must be completed and fully embedded into maintenance workflows and solutions prior to Handover. Maintenance software and systems should be populated, and staff provided with information they need in a form that enables them to understand the operation and the tasks they must perform to safely and efficiently use and care for the facility. This information includes, for example:

A. HAZOP and other safety and environmental analyses with associated documentation, strategies for

inspection, training, certification, etc.

B. Control system, historian, and alarm system configuration information (hardware addresses, conversion parameters, loop parameters, etc.).

C. Reliability analysis for all critical equipment including all information and strategies for inspection and preventive maintenance (PM).

D. Material masters and supplier data bases for direct and maintenance, repair and operations materials fully initialized in the facility's procurement software.

E. Operating, inspection, and maintenance procedures.

5.2.4. Successful Outcomes:

- ✓ **Effective information handover process**– Stakeholders clearly articulate the kinds of information the Airport needs and how it will use that information to operate and maintain the facility.

- ✓ **Operation & Maintenance information embedded into workflows** – provide O&M systems and staff with information they need in a form that enables them to understand the operation and the tasks they must perform to safely, efficiently use and care for the facility.

5.3. Deliver Training

Effective training delivery is key to a successful transition from a current to future-state. For a new facility to operate effectively, staff must be confident in their roles and able to respond in any situation. The tracking and reporting of overall readiness of critical operational staff is essential to ensure the smooth execution and reliability of Operational Exercises and overall staff readiness for Opening Day. The following approach will be used to ensure all training and familiarization activities are scheduled, delivered, tracked, and progress reported to the OSC.

Primary Responsibility	PM/ORAT
Key Collaborator(s)	AM & Mx
Project Tier Applicability	All

5.3.1. Step 1: Coordinate and Execute Training and Familiarization

The ORAT Team should coordinate with stakeholders as early as possible to deliver against the Training and Familiarization Schedule. Considerations for attendance include role prioritization, shift patterns, and inclusion as a participant in the Operational Exercises. Leaders from stakeholder groups will need to play an active role in identifying staff and ensuring their presence at sessions.

The activities and sessions should be delivered in line with the Training and Familiarization Schedule and the materials developed earlier in the ORAT Program. At each session, attendance should be recorded via attendance forms or Workday, and feedback received.

5.3.2. Step 2: Track People Readiness Progress

The ORAT Team should track people readiness progress via recorded attendance. Where training and familiarization is not overseen by the ORAT Team, such as with technical training, they should receive attendance registers from the Trainer. A People Readiness Tracker should be established to consolidate and monitor the percentage of completion and overall readiness of critical operational staff. This tracker is developed and maintained in consultation with each of the stakeholders who have an operational resource within a facility. By developing this tool early, it will allow for tracking of individual stakeholder progress and a clear status for reporting overall program readiness. The Trainer should proactively monitor employee performance, acceptance, participation rates and gather participant feedback across all Training and Familiarization activities delivered to stakeholders.

Training and Familiarization activities may continue long after Opening Day due to the continual onboarding of new staff. Trainers will continue to collect training delivery data post to evaluate the ongoing effectiveness.

5.3.3. Successful Outcomes:

- ✓ **Training assurance and staff readiness** - upon completion of Training and Familiarization, staff are prepared to demonstrate proficiency during the Operational Exercises.

5.4. Deliver Operational Exercises

Primary Responsibility	ORAT
Key Collaborator(s)	Ops, Mx, PM
Project Tier Applicability	All

The delivery of Operational Exercises is dependent on functioning systems and equipment, trained staff, and having unimpeded access to relevant building areas. Once these prerequisites are met, Operational Exercises can be performed with the intent to identify and resolve any issues with systems, processes, or training in a controlled environment. An Operational Exercise is complete when the exercise has been carried out, participants, volunteers, and observers have been debriefed and all the Operational Exercise results and feedback have been collected. The following approach will be used to ensure all Operational Exercises are appropriately scheduled and delivered for the project.

5.4.1. Step 1: Develop Controller and Evaluator Handbook

The logistic and resource requirements for each Operational Exercise will depend on the complexity of the Exercise, the areas in scope, and the number of volunteers participating. Tabletops and Drills may not require as much logistics support (e.g., providing Operational Exercise materials such as scripts may be all that is required). Functional and Full-scale Exercises may require significantly more logistics and resource support (e.g., preparing staging areas for volunteers, catering, volunteer incentives, bussing/transportation and publicity and communications support).

A Controller and Evaluator Handbook, developed by the ORAT Team, is a step-by-step script detailing what is required from Operational Exercise volunteers and participants to ensure the objective is met. This document will detail the run-of-show and key responsibilities during the delivery of the exercise. It provides guidance for participants and instructions on how to successfully take part in the exercise to achieve the testing of a particular process, function, or facility.

5.4.2. Step 2: Monitor Status of all Dependencies

For an Operational Exercise to be delivered, all dependencies (e.g., training delivery and facilities and systems activation), should be complete. The ORAT Team should collaborate with the Project Team to understand the status of all dependency items and, if needed, adjust the schedule to allow the Exercise to be delivered.

5.4.3. Step 3: Schedule a Go/No-Go with the OSC

Once the Operational Exercises have been scheduled, and the Controller and Evaluator Handbook completed, the ORAT Project Lead should schedule a Go/No-Go with the OSC 48/24 hours prior to the Operational Exercise date for final review of all dependencies and potential conflicts. Before commencing each Operational Exercise, assessments will be made to:

A. Establish that all dependencies have been met and if not, assess whether approved workarounds are in place;

B. Equipment and systems have been confirmed to be functional;

C. Confirm necessary resources (people, equipment, site access) are available; and

D. Confirm that all logistics are in place.

A Go/No-Go decision is made based on these assessments and clearly defined Go/No-Go criteria before executing any exercise. If a “Go” decision is made, the Operational Exercise will be executed in accordance with the agreed upon Controller and Evaluator Handbook. If a “No-Go” decision is made, the Operational Exercise may be postponed to a later date, the project risk register needs to be updated, and a Go-Live risk assessment completed for each failed process.

5.4.4. Step 4: Execute Operational Exercises

Operational Exercises are executed in accordance with the established Controller and Evaluator Handbook and

the agreed schedule. Operational Exercises commence with Drills, which test a subset of the overall operations process and allow readiness assessments to be made on a small scale to confirm the effectiveness of the subset process. This then progresses to Functional exercises, which confirm, on a small scale, the effectiveness of the full end-to-end process. Optionally, the Operational Exercise plan may include a series of exercises to assess fall back and emergency scenarios, which assess emergency processes. Culminating in Full-scale exercises, which are a combination of drills to test combined system operation, end-to-end processes, and integrated operational scenarios.

5.4.5. Step 5: Collect Observer Feedback

Critical to Exercise delivery is capturing feedback using prepared feedback forms received from exercise volunteers and observers. They offer the opportunity for open-ended feedback and are easily analyzed and collated post-operational exercise execution. Specific observer feedback forms for select SME observers may be created to collect observations and feedback regarding a specific function, process, or feature.

5.4.6. Step 6: Compile an Exercise After Action Review (AAR) and record issues raised by Operational Exercise in the Issues Tracker

This is the concluding phase in the Operational Exercise process and its main output is the AAR. After each Exercise, an analysis is performed to review participant feedback to determine if the Operational Exercise objectives and exercise evaluation criteria were met. An AAR is produced as an accurate record of outcomes of the Exercise, and details, whether the Exercise passed or failed, issues identified and prioritized, and action items assigned. The results of the Operational Exercise will be reviewed and evaluated to determine the necessity of:

A. Corrective action (captured in the Issues Tracker).

B. Process changes or new fallback procedures.

C. Re-running an Operational Exercise, or portion of an Exercise.

5.4.7. Successful Outcomes:

- ✓ **Demonstrated Operational Readiness** – confirmation of system or asset implementation, updated processes, and staff training.

- ✓ **Issues identified and assigned for resolution** – issues captured through observation and feedback are assigned or escalated to the appropriate stakeholders for resolution.

5.5. Mobilizations & Space Handover

Primary Responsibility	Move Coordinator
Key Collaborator(s)	ORAT
Project Tier Applicability	All

Whether the Airport is embarking on a completely new build or simply shuffling operational and tenant space arrangements, move management is a core component of the Transition and Cutover Plan. Even a small move can consist of many connected parts that need to be synchronized, and it's the collective job of the ORAT Team, move coordinator, and facility managers team to make sure that the day-to-day operations of the Airport are not heavily impacted in the process. If stakeholders are not properly looped in and all the variables considered, a move can quickly cost more and take longer than anticipated. The following approach will be used to ensure Mobilization and Space Handover is appropriately delivered for the project.

5.5.1. Step 1: Onboard Move Coordinator and Prepare Moves

The ORAT Team and/or Project Team should identify or onboard a Move Coordinator when it becomes necessary to relocate people, services, and equipment on a larger scale. The Move Coordinator should be responsible for planning, scheduling, and coordinating the logistics of the relocation.

The ORAT Team and Move Coordinator will collaborate with stakeholders to schedule tenant walkthroughs to verify space, flag issues, and identify furniture, equipment, or service locations. Issues with the space may be escalated as needed, and furniture/equipment locations may be collated and relayed to the move specialists.

5.5.2. Step 2: Execute Space Handover and Moves

The ORAT Team and Move Coordinator should execute stakeholder moves; provide support during stakeholder relocations including advice regarding security procedures, access, and pass functionality as well as delivery of goods. Move communication collateral should be created to support the delivery of critical information and key messages regarding the move, including but not limited to:

A. Move schedule e.g., move date, packing and unpacking dates;

B. Changes to operation during move-in;

C. Available move support; and

D. Contact information for FM, IT, etc.

After the move is complete and the relocated staff becomes familiar with the space, the Move Coordinator shall follow-up with the stakeholder's point of contact to collect any post-move punch list items to be completed by appropriate parties.

The methods and frequency of communication delivered to each stakeholder should be appropriate to the Project Tier.

5.5.3. Successful Outcomes:

- ✓ **Minimal disruptions to operations** – through planned and coordinated moves, ideally executed during off-peak hours or seasons if possible.
- ✓ **Minimizing remedial work post space moves** – through tenant walkthroughs to verify space, flag issues, and identify furniture, equipment, or service locations.

5.6. Transition & Cutover

Primary Responsibility	ORAT
Key Collaborator(s)	PM
Project Tier Applicability	All

The final activity in the Deliver Phase, Execute, is to verify the

completion of construction and planned ORAT activities and deliver transition and cutover. The following approach will be used to ensure operational readiness at activation and a smooth transition into operations.

5.6.1. Step 1: Execute various Transition Checklist Items

The Transition Readiness Checklist, developed earlier in the ORAT Program, details the activities needed for operations groups to be ready to inherit the new systems and/or spaces. The ORAT Team should facilitate the responsible owners of checklist items to ensure they are completed in a timely manner. A regular meeting can aid in ensuring owners are held accountable to completing their items.

5.6.2. Step 2: Schedule a Go/No-Go with the OSC

Vital to the success of the transition process, and the final countdown to opening, is a Go/No-Go decision point on the opening date. Validation checks on the proposed opening date should be made progressively throughout the project with key milestones highlighted in the overall ORAT program to assess the viability of the proposed opening date. As the project advances, the ORAT Team should conduct regular checks to evaluate the likelihood of achieving the target Opening Day. A risk assessment should be performed, considering key indicators such as construction completion, training and familiarization completion, and operational exercise completion. The target Opening Day should be regularly reviewed against the current perceived risk.

A. If, at the time of assessment, the risk is perceived to be high and sufficient mitigation cannot be put in place to reduce the risk, the ORAT Steering Committee should recommend to the Executive Leadership whether to delay the Opening Day.

B. If, at the time of assessment, the risk is perceived as medium or low, with sufficient mitigation in place to reduce the risks, then the Opening Day should be endorsed by the OSC, and the project should proceed with transition plans towards the final countdown.

The final assessment for activation should be made at the 30-day countdown period (at the very latest), considering all measures as well as the results of the 90- and 60-day countdown process. The same assessment practice will be followed to give a final Go/No-Go decision for the Opening Day. The status of key indicators for activation and risk mitigation plans should be communicated to stakeholders as needed. Key issues that may affect the activation date include:

A. Construction progress;

B. System installation & commissioning progress;

C. Acceptance of facilities and systems;

D. Stakeholder readiness;

E. Recruitment, training and familiarization progress;

F. Unresolved critical issues raised during Operational Exercises; and

G. Achievement of statutory certification.

Step 3: Support Go-Live

The final countdown to opening is the most critical period, marked by intense activities and numerous relocations and moves. A clear central command and control process is crucial for success during this period. This process should monitor and manage the moves, resolve issues as they arise, and mitigate risks as needed.

Large transition projects often require a task force to coordinate the activation and go-live time period. Their mission is to coordinate the technical, operations, and procurement teams and keep stakeholders informed of

the project status, make cross-functional decisions, and, crucially, resolve operational complications.

The task force should be composed of employees familiar with the transition project scope, assigned resources, and task schedule so they can manage the transition and coordinate on dealing with any issues. These employees should be identified at least a month in advance to ensure they are prepared for go-live. Over the go-live period, this task force will have overall control of the transition process and ensure any activities to deliver it are coordinated and executed.

Most facility activations include a launch event. While arranging this event should not be the responsibility of the ORAT Team, close coordination is necessary to ensure the event's timing is appropriate and does not impact opening and readiness activities. The timing of the launch event and the actual opening date should not coincide; previous experience has shown that launch events can occur either before the Opening Day (typically with a soft launch or long migration) or after the Opening Day with an 'official opening' celebrating a successful transition. By coordinating the event around a successful ORAT Program, a positive public relations message can be presented without impacting operational readiness preparations and activities.

5.6.3. Step 4: Identify post-opening support requirements

Once the ORAT Program has been successfully completed, and the facility has been opened to the public, core members of the ORAT Team are recommended to perform a monitoring and evaluation function to determine if the facility is operating as expected, including the following reviews:

A. coordination of any remedial works required;

B. handover of all ORAT documentation; and

C. recommendations to improve facilities or operational processes related to facilities and/or equipment.

5.6.4. Successful Outcomes:

- ✓ **Opening Assurance** – by verifying the completion of construction and planned ORAT activities, and support activation.

9.2 Deliver Phase: Execute Toolkit

Checklist	Templates	Reference Resources
☐ Asset Information ☐ Critical Spares List ☐ Maintenance Plans ☐ Handover Plans for assets and systems ☐ People Readiness Tracker ☐ Operational Exercise Documents ☐ Controller and Evaluator Handbook ☐ Go/No-Go Meetings ☐ Observer Feedback ☐ Exercise After Action Review	[Controller and Evaluator Handbook template] [ORAT Exercise After Action Review template]	[Refer to the Project Management Manual and Project Life Cycle (PLC)]

End of Chapter

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Chapter 6- Closeout Phase

6.0. Introduction

Purpose: To provide on-going support for issues resolution and ensuring changes are embedded and there is a sustained operation.

Considerations:

A. Have we celebrated our successes?

B. Who is responsible for Issues Management & Resolution?

C. How do we ensure performance is monitored?

D. How will learning be embedded to improve our next project?

6.1. Issue Management & Resolution

Issue Management as previously defined is the process of identifying, analyzing, resolving, and preventing problems that can arise during project implementation. The following approach will be used to ensure issues are properly captured and actioned by stakeholders.

Primary Responsibility	ORAT
Key Collaborator(s)	Ops, Mx, PM
Project Tier Applicability	All

6.1.1. Step 1: Update ORAT Issues Tracker

The ORAT Team should actively be utilizing the Issues Tracker, developed during the Define Phase and used throughout the project to compile issues raised throughout the delivery, and specifically issues observed and raised during each Operational Exercise and on Opening Day. The ORAT Project Lead should work with stakeholders to ensure they take ownership and report their open issues and track their resolution.

Issues should be tracked by the ORAT Project Lead until 1-month post-Go-Live. Following that, Issues should be turned over to relevant Stakeholder Groups to continue through their own resolution processes.

6.1.2. Successful Outcomes:

- ✓ **Clearly defined issues resolution ownership** – of activation and "Day 2" issues.

6.2. Conduct Lessons Observed Review

Primary Responsibility	PM
Key Collaborator(s)	ORAT
Project Tier Applicability	All

Learning occurs on every project. We learn from our own project experiences as well as the experiences of others. Project managers, team members and leadership should all participate in the lessons observed sessions, review the outcomes and make decisions on how to use the knowledge gained. Sharing lessons learned among project team members prevents an organization from repeating the same mistakes and allows them to take advantage of organizational best practices. Innovative approaches and good work practices can be shared with others. Lessons observed can be used to improve future projects and future stages of current projects. The following approach will be used to ensure lessons learned are captured and stored for reference on future projects.

6.2.1. Step 1: ORAT Support in a Lessons Observed Survey and/or Workshop

A lessons learned workshop focuses on identifying project success and project failures and includes recommendations to improve future performance on projects. If a lessons learned workshop is conducted for the project, the ORAT Project Lead will participate in the discussion and provide insights to apply on future projects. ORAT Project Lead will facilitate a separate workshop for end users collecting their input on the projects and operational lessons learned through a survey. The Lessons Observed should be documented by the Project Team in a format that allows for reference and continuous improvement.

6.2.2. Successful Outcomes:

- ✓ **Continuous Learning** – from lessons learned to improve future projects and future stages of current projects.

9.3 Close Out Phase: Toolkit

Checklist	Templates	Reference Resources
☐ Issues Tracker ☐ Lessons Observed Workshop ☐ Consolidated Lessons Observed Findings		[Refer to the Project Management Manual and Project Life Cycle (PLC)]

End of Chapter

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